



Grading Scheme

Quiz 1 : 10%

Quiz 2 : 10%

Assignments: 10%

Midsem : 25%

Endsem : 50%

Newtonian Mech → Inconsistent with Maxwell's Eq^{ns}.

↓
speed of light is const.

∴ STR is needed where speed of light is constant.

In STR → Gravity is not compatible with inertial reference frames.

Gravity is not invariant with change in inertial reference frames.

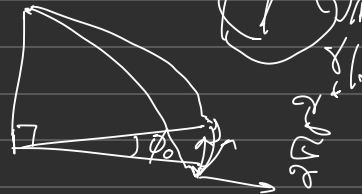
↓
General Relativity is needed!



$$\int_0^{\alpha} (R \sin \theta d\phi) (R d\theta)$$

$$\underline{R^2 \alpha = A}$$

$$\sum = \pi \times \alpha$$



$$\frac{\alpha}{2\pi} = \frac{A}{(\pi R^2 / 2)}$$

Lecture 2 (8 Jan)

Euler-Lagrange Eqⁿ -

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} = 0$$

$$L = T - V$$

$$\text{Action: } S = \int_{t_A}^{t_B} L dt$$

$$L = \frac{1}{2} m \dot{x}^2 - V(x)$$

$$\hookrightarrow \frac{d(m\dot{x})}{dt} + \frac{dV}{dx} = 0 \Rightarrow m\ddot{x} = -\frac{dV}{dx} = F_{\text{ext}} \quad [2^{\text{nd}} \text{ Law}]$$

Coordinate frame

We have a curvilinear surface over which the motion is constrained.

Let coordinates be (q_1, q_2)

Let $d\vec{s}$ be displacement vector.

Let $\vec{e}_1, \vec{e}_2$ be basis vectors.

$$d\vec{s} = dq^1 \vec{e}_1 + dq^2 \vec{e}_2 = \sum_{i=1}^2 dq^i \vec{e}_i = dq^i e_i \quad (\text{Einstein summation convention})$$

$$ds^2 = (dq^i \vec{e}_i) \cdot (dq^j \vec{e}_j) = dq^i dq^j \underbrace{g_{ij}}_{\text{Metric Tensor}}$$

$$[g_{ij}] = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} ; (g_{ij} = \vec{e}_i \cdot \vec{e}_j) \quad \hookrightarrow \text{Metric Tensor}$$

Metric Tensor
is symmetric

$$\text{i.e. } g_{ij} = g_{ji}$$

Lagrangian:

$$L = T - V = T \quad (\because V=0)$$

$$= \frac{1}{2} m v^2 = \frac{1}{2} m \left(\frac{d\vec{s}}{dt} \cdot \frac{d\vec{s}}{dt} \right) = \frac{1}{2} m \left(\frac{dq^i}{dt} \frac{dq^j}{dt} e_i \cdot e_j \right)$$

$$L = \frac{1}{2} m g_{ij} \frac{dq^i}{dt} \frac{dq^j}{dt}$$

Euler-Lagrange eqⁿ -

$$\frac{d}{dt} \left[\frac{\partial L}{\partial \dot{q}_k} \right] - \frac{\partial L}{\partial q_k} = 0 \quad k \in \{1, 2\}$$

$$\frac{\partial L}{\partial \dot{q}_k} = \left[\frac{1}{2} m g_{ij} \delta_k^i \dot{q}^j + \frac{1}{2} m g_{ij} \dot{q}^i \delta_k^j \right] = m g_{ki} \dot{q}^i$$

$$\begin{aligned} \frac{d}{dt} (m g_{ki} \dot{q}^i) &= m \frac{dg_{ki}}{dt} \dot{q}^i + m g_{ki} \ddot{q}^i \\ &= m \frac{\partial g_{ki}}{\partial q_j} \dot{q}^i \dot{q}^j + m g_{ki} \ddot{q}^i \end{aligned}$$

$$\frac{\partial L}{\partial q_k} = \frac{1}{2} m \frac{\partial g_{ij}}{\partial q_k} \dot{q}^i \dot{q}^j$$

$$\therefore m \frac{\partial g_{ki}}{\partial q_j} \dot{q}^i \dot{q}^j + m g_{ki} \ddot{q}^i - \frac{1}{2} m \frac{\partial g_{ij}}{\partial q_k} \dot{q}^i \dot{q}^j = 0$$

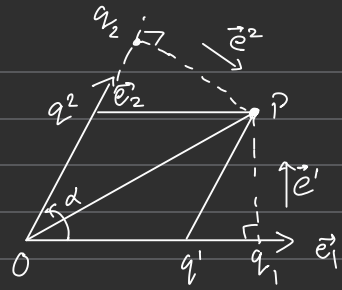
$$\Rightarrow \boxed{\frac{1}{2} \left[\frac{\partial g_{ki}}{\partial q_j} + \frac{\partial g_{kj}}{\partial q_i} - \frac{\partial g_{ij}}{\partial q_k} \right] \dot{q}^i \dot{q}^j + g_{ki} \ddot{q}^i = 0} \quad \text{Equation of Motion}$$

$$q_1 = q^1 + q^2 \cos \alpha$$

$$q_2 = q^2 + q^1 \cos \alpha$$

$$\therefore \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = \begin{bmatrix} 1 & \cos \alpha \\ \cos \alpha & 1 \end{bmatrix} \begin{pmatrix} q^1 \\ q^2 \end{pmatrix}$$

$$\longrightarrow \boxed{q_i = g_{ij} q^j}$$



$\vec{e}_i$: tangent basis
(covariant basis)

$\vec{e}^i$: cotangent basis
(contravariant basis)

$$d\vec{s} = dq^i \vec{e}_i = dq_i \vec{e}^i$$

$$ds^2 = dq^i dq^j g_{ij} = dq_i dq_j g^{ij}$$

Lecture 3 (12 Jan)

$$\frac{1}{2} \left[\frac{\partial g_{ki}}{\partial q_j} + \frac{\partial g_{kj}}{\partial q_i} - \frac{\partial g_{ij}}{\partial q_k} \right] \dot{q}^i \dot{q}^j + g_{ki} \ddot{q}^i = 0$$

$$\rightarrow \frac{g^{lk}}{2} \left[\frac{\partial g_{ki}}{\partial q_j} + \frac{\partial g_{kj}}{\partial q_i} - \frac{\partial g_{ij}}{\partial q_k} \right] \dot{q}^i \dot{q}^j + g^{lk} g_{ki} \ddot{q}^i = 0$$

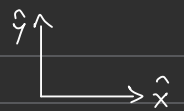
$$\rightarrow \frac{1}{2} g^{lk} \left[\frac{\partial g_{ki}}{\partial q_j} + \frac{\partial g_{kj}}{\partial q_i} - \frac{\partial g_{ij}}{\partial q_k} \right] \dot{q}^i \dot{q}^j + \ddot{q}^l = 0$$

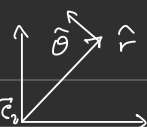
Γ_{ij}^l

$(dq_i = g_{ij} dq^j = g_{ij} g^{jk} dq_k = \delta_i^k dq_k)$
Christoffel Symbol
of 2nd Kind.

$$\rightarrow \boxed{\ddot{q}^l + \Gamma_{ij}^l \dot{q}^i \dot{q}^j = 0}$$

Flat 2D surface.

① $x-y$  $\vec{e}_1 = \hat{i}$
 $\vec{e}_2 = \hat{j}$

② Plane Polar  $\vec{r} = r \cos \theta \vec{e}_1 + r \sin \theta \vec{e}_2$

$$\vec{e}_r = \frac{\partial \vec{r}}{\partial r} = \cos \theta \vec{e}_1 + \sin \theta \vec{e}_2$$

$$\vec{e}^r = \nabla_x r = \cos \theta \vec{e}_1 + \sin \theta \vec{e}_2$$

$$\vec{e}_\theta = \frac{\partial \vec{r}}{\partial \theta} = -r \sin \theta \vec{e}_1 + r \cos \theta \vec{e}_2$$

$$\vec{e}^\theta = \nabla_x \theta = -\frac{1}{r} \sin \theta \vec{e}_1 + \frac{1}{r} \cos \theta \vec{e}_2$$

$$g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} \quad g^{ij} = \begin{pmatrix} 1 & 0 \\ 0 & 1/r^2 \end{pmatrix}$$

Method 1

$$\Gamma_{ij}^k g_k = \frac{\partial g_i}{\partial q^j} \Rightarrow \left[\Gamma_{ij}^k = g^k \cdot \frac{\partial g_i}{\partial q^j} \right]$$

$$\Gamma_{\theta\theta}^r = \vec{e}^r \cdot \left(\frac{\partial \vec{e}_\theta}{\partial \theta} \right) = (\cos \theta \vec{e}_1 + \sin \theta \vec{e}_2) \cdot (-r \cos \theta \vec{e}_1 - r \sin \theta \vec{e}_2)$$

$$= -r \cos^2 \theta - r \sin^2 \theta = -r$$

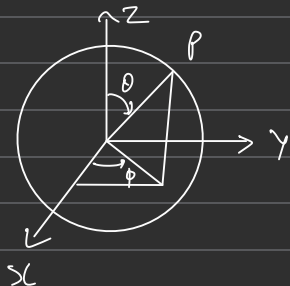
Method 2

$$\Gamma_{\theta\theta}^r = \frac{1}{2} g^{rr} \left[\frac{\partial g_{r\theta}}{\partial \theta} + \frac{\partial g_{\theta r}}{\partial \theta} - \frac{\partial g_{\theta\theta}}{\partial r} \right] + \frac{1}{2} g^{r\theta} \left[\frac{\partial g_{\theta\theta}}{\partial \theta} + \frac{\partial g_{\theta\theta}}{\partial \theta} - \frac{\partial g_{\theta\theta}}{\partial \theta} \right] = -r$$

||ly, $\Gamma_{rr}^r = 0$, $\Gamma_{rr}^\theta = 0$, $\Gamma_{r\theta}^r = 0$, $\Gamma_{r\theta}^\theta = \frac{1}{r}$, $\Gamma_{\theta\theta}^r = -r$, $\Gamma_{\theta\theta}^\theta = 0$

$\therefore$ Eqⁿ of Motion: $\boxed{\begin{array}{l} \text{① } \ddot{r} - r\dot{\theta}^2 = 0 \\ \text{② } \ddot{\theta} + \frac{2}{r}\dot{r}\dot{\theta} = 0 \end{array}}$

Surface of a Sphere



$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{x^2 + y^2}}{z} \right)$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

$$\vec{e}_\theta = r \cos \theta \cos \phi \vec{e}_1 + r \cos \theta \sin \phi \vec{e}_2 - r \sin \theta \vec{e}_3$$

$$\vec{e}_\phi = -r \sin \theta \sin \phi \vec{e}_1 + r \sin \theta \cos \phi \vec{e}_2$$

$$[g_{ij}] = \begin{bmatrix} r^2 & 0 \\ 0 & r^2 \sin^2 \theta \end{bmatrix} \quad [g^{ij}] = \begin{bmatrix} \frac{1}{r^2} & 0 \\ 0 & \frac{1}{r^2 \sin^2 \theta} \end{bmatrix}$$

$$\nabla \vec{e}_\theta = \begin{pmatrix} -r \sin \theta \cos \phi \vec{e}_1 - r \sin \theta \sin \phi \vec{e}_2 - r \cos \theta \vec{e}_3 \\ -r \cos \theta \sin \phi \vec{e}_1 + r \cos \theta \cos \phi \vec{e}_2 \end{pmatrix}$$

$$\nabla \vec{e}_\phi = \begin{pmatrix} -r \cos \theta \sin \phi \vec{e}_1 + r \cos \theta \cos \phi \vec{e}_2 \\ -r \sin \theta \cos \phi \vec{e}_1 - r \sin \theta \sin \phi \vec{e}_2 \end{pmatrix}$$

$$\begin{aligned} \vec{e}^\theta &= \cos \theta \cos \phi \vec{e}_1 + \cos \theta \sin \phi \vec{e}_2 - \sin \theta \vec{e}_3 \\ \vec{e}^\phi &= \frac{-\sin \phi}{r \sin \theta} \vec{e}_1 + \frac{\cos \phi}{r \sin \theta} \vec{e}_2 \end{aligned}$$

$$\Gamma_{\theta\theta}^\theta = 0$$

$$\Gamma_{\theta\theta}^\phi = 0$$

$\therefore$ Equations of Motion :-

$$\Gamma_{\theta\phi}^\theta = 0$$

$$\Gamma_{\theta\phi}^\phi = \cot \theta$$

$$\textcircled{1} \ddot{\theta} - \sin \theta \cos \theta \dot{\phi}^2 = 0$$

$$\Gamma_{\phi\theta}^\theta = 0$$

$$\Gamma_{\phi\theta}^\phi = \cot \theta$$

$$\textcircled{2} \ddot{\phi} + 2 \cot \theta \dot{\theta} \dot{\phi} = 0$$

$$\Gamma_{\phi\phi}^\theta = -\sin \theta \cos \theta$$

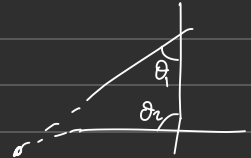
$$\Gamma_{\phi\phi}^\phi = 0$$

Lecture 4 (15 Jan)

Euclid's Postulates -

- ① Any 2 points can be connected via a straight line.
- ② Any straight line can be extended indefinitely along both of its end points.
- ③ Circle is a collection of points equidistant from a fixed point.
- ④ All right angles are congruent.
- ⑤ $\theta_1 + \theta_2 < 180^\circ \Rightarrow$ Point of Intersection exists.

Does not hold in Non-Euclidean Geometry.



Curvature

finding maximum & minimum radius of curvatures at a point on a cylindrical surface.



$$r_{\min} = R$$

$$r_{\max} \rightarrow \infty$$

$$\therefore K = \frac{1}{r_{\min}} \times \frac{1}{r_{\max}}$$

Gaussian Curvature -

Extrinsic Definition.

[we have to come out of surface to define $r_{\min}, r_{\max}$]

$$K = \lim_{r \rightarrow 0} \frac{3}{\pi} \left(\frac{2\pi r - C(r)}{r^3} \right)$$

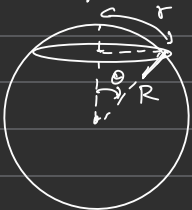
Intrinsic Definition.

[we can calculate curvature while remaining inside the surface]

↳ Here, r is radius of circle along the surface.

$C(r)$ is the circumference of the circle of radius r .

Ex. Sphere -



$$r = R\theta$$

$$C(r) = 2\pi R \sin \theta = 2\pi R \sin\left(\frac{r}{R}\right)$$

$$= 2\pi R \left[\frac{r}{R} - \left(\frac{r}{R}\right)^3 \frac{1}{6} + \dots \right]$$

$$= 2\pi r - \frac{\pi r^3}{3R^2} + \dots$$

$$\therefore K = \lim_{r \rightarrow 0} \frac{3}{\pi} \left(\frac{2\pi r - C(r)}{r^3} \right) = \frac{1}{R^2}$$

==>

η = Separation b/w 2 longitudes.

$$\eta = R \phi \sin \theta$$

$$\eta = R \phi \sin\left(\frac{s}{R}\right)$$

$$\frac{d\eta}{ds} = \phi \cos\left(\frac{s}{R}\right)$$

$$\frac{d^2\eta}{ds^2} = -\frac{\phi}{R} \sin\left(\frac{s}{R}\right) = -\frac{\eta}{R^2} = -K\eta$$

$K \equiv \frac{1}{R^2}$: Gaussian Curvature



$$\boxed{\frac{d^2\eta}{ds^2} = -K\eta}$$

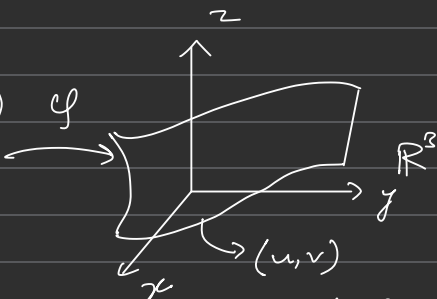
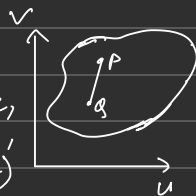
Geodesic Deviation.

↳ This eqⁿ holds for any Real 2D Surface.

Ist fundamental form:

$$P: (u_0, v_0) \xrightarrow{\gamma} (x_0, y_0, z_0)$$

$$Q: (u_0 + du, v_0 + dv) \xrightarrow{\gamma} (x_0 + dx, y_0 + dy, z_0 + dz)$$



$$\begin{aligned} x_0 + dx &= x(u_0 + du, v_0 + dv) \\ &= x_0 + \frac{\partial x}{\partial u} du + \frac{\partial x}{\partial v} dv + O(du^2) \end{aligned}$$

$$\begin{aligned} x &\equiv x(u, v) \\ y &\equiv y(u, v) \\ z &\equiv z(u, v) \end{aligned}$$

$$ds^2 = dx^2 + dy^2 + dz^2$$

$$\hookrightarrow ds^2 = E du^2 + 2F du dv + G dv^2 \quad \} \text{ Ist Fundamental Form}$$

$$E = \left(\frac{\partial x}{\partial u}\right)^2 + \left(\frac{\partial y}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial u}\right)^2$$

$$G = \left(\frac{\partial x}{\partial v}\right)^2 + \left(\frac{\partial y}{\partial v}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2$$

$$F = \left(\frac{\partial x}{\partial u}\right)\left(\frac{\partial x}{\partial v}\right) + \left(\frac{\partial y}{\partial u}\right)\left(\frac{\partial y}{\partial v}\right) + \left(\frac{\partial z}{\partial u}\right)\left(\frac{\partial z}{\partial v}\right)$$

Alternate method -

$$\vec{r}(u,v) = x(u,v)\hat{i} + y(u,v)\hat{j} + z(u,v)\hat{k}$$

$$\vec{r}_u = \frac{\partial \vec{r}}{\partial u}, \quad \vec{r}_v = \frac{\partial \vec{r}}{\partial v}$$

$$E = \vec{r}_u \cdot \vec{r}_u; \quad F = \vec{r}_u \cdot \vec{r}_v; \quad G = \vec{r}_v \cdot \vec{r}_v$$

$$[g] = \begin{bmatrix} E & F \\ F & G \end{bmatrix}; \quad \boxed{I(u,v) = E du^2 + 2F du dv + G dv^2}$$

Lecture 5 (19 Jan)

Lorentz Transformation

$$x' = \gamma (x - vt)$$
$$t' = \gamma \left(t - \frac{vx}{c^2} \right)$$



$$\gamma = \frac{1}{\sqrt{1 - (v/c)^2}}$$

Assignment - 1

Q1.

$$\frac{dx'}{dt'} = \gamma \left(\frac{dx}{dt} - v \right) \frac{dt}{dt'}$$

$$\frac{dt'}{dt} = \gamma \left(1 - \frac{v}{c^2} \frac{dx}{dt} \right)$$

$$u' = \frac{\frac{dx}{dt} - v}{1 - \frac{v}{c^2} \frac{dx}{dt}}$$

$$\Rightarrow \frac{dx}{dt} \left(-\frac{vu'}{c^2} \right) + u' = \frac{dx}{dt} - v$$

$$\Rightarrow \boxed{\frac{dx}{dt} = \frac{v + u'}{1 + \frac{vu'}{c^2}}}$$

$$Q2. \quad \frac{dy}{dt} = u_y \rightarrow \frac{dy'}{dt'} = \frac{dy}{dt} \frac{dt}{dt'} = \frac{u_y}{\gamma(1 - \frac{v}{c^2} u_x)}$$

$$u'_x = -v \quad ; \quad u_x = 0$$

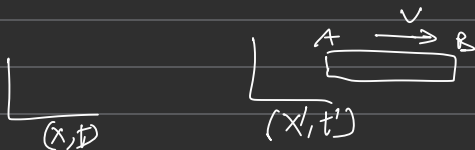
$$u'_y = u_y \sqrt{1 - \left(\frac{v}{c}\right)^2}$$

$$Q3. \quad t' = \gamma \left(t - \frac{vx}{c^2} \right) = \gamma t \quad \begin{matrix} S \\ (0,0) \end{matrix}, \begin{matrix} S' \\ (0,0) \end{matrix}$$

$$x' = \gamma (x - vt) = -\gamma vt \quad \begin{matrix} (0, t) \\ (-vt', t') \end{matrix}$$

$$= -vt'$$

$$Q4. \quad L_0 = \gamma L \rightarrow L = \frac{L_0}{\gamma} < L_0$$



$$Q5. \quad \Delta s^2 = -c^2 \Delta t^2 + \Delta x^2$$

$$\Delta s'^2 = -c^2 \Delta t'^2 + \Delta x'^2$$

$$= -c^2 \gamma^2 \left(\Delta t - \frac{v \Delta x}{c^2} \right)^2 + \gamma^2 (\Delta x - v \Delta t)^2$$

$$= -c^2 \gamma^2 \left(\Delta t^2 - \frac{2v \Delta x \Delta t}{c^2} + \frac{v^2 \Delta x^2}{c^4} \right) + \gamma^2 \left(\Delta x^2 - 2v \Delta t \Delta x + v^2 \Delta t^2 \right)$$

$$= -c^2 \gamma^2 \left[\Delta t^2 - \frac{2v \Delta x \Delta t}{c^2} + \frac{v^2 \Delta x^2}{c^4} - \frac{\Delta x^2}{c^2} + \frac{2v \Delta t \Delta x}{c^2} - \frac{v^2 \Delta t^2}{c^2} \right]$$

$$= -c^2 \gamma^2 \left[\frac{\Delta t^2}{\gamma^2} - \frac{\Delta x^2}{c^2 \gamma^2} \right]$$

$$= -c^2 \Delta t^2 + \Delta x^2$$

$$= \underline{\Delta s^2}$$

Invariant

Q6. $v = 0.99c$ $\tau_0 = 1.77 \times 10^{-8} \text{ s}$
 $\gamma = 7.09$

$$\begin{matrix} (0,0) \\ (0,0) \end{matrix} \rightarrow \begin{matrix} (x,t) \\ (0,\tau_0) \end{matrix}$$

$$x = \gamma(0 + v\tau_0) \approx 37.24 \text{ m}$$

$$t = \gamma\left(\tau_0 + \frac{v(0)}{c^2}\right) \approx 1.2 \times 10^{-7} \text{ sec}$$

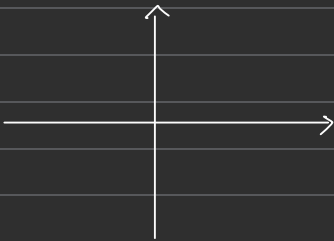
$\therefore$ Results are consistent -

Q7.

(a) $v = \frac{100}{5 \times 10^{-6}} = 2 \times 10^7 \text{ m/s}$

(b) $t_0 = \frac{t_A}{\gamma} = 4.99 \times 10^{-6} \text{ sec}$

Q8.



$$\gamma = \frac{5}{3}$$

$$\vec{r} = (ct)(\cos\theta \mathbf{e}_1 + \sin\theta \mathbf{e}_2)$$

$$\vec{v} = \frac{4}{5}c \mathbf{e}_1$$

$$x = \gamma(x' + vt')$$

$$t = \gamma\left(t' + \frac{4x'}{5}\right)$$

$$y = ct' \sin\theta'$$

$$x = \frac{5}{3} \left(ct' \cos\theta' + \frac{4}{5} ct' \right)$$

$$\boxed{x = \frac{ct'}{3} (5\sqrt{1-y'^2} + 4)}$$

$$= \frac{ct'}{3} (5\cos\theta' + 4)$$

$$\gamma = 5/3$$

Q9. $x = 16 \text{ light-year}$

$$x' = \gamma(x - vt) = 0$$

$$t = \frac{5x}{4c}$$

$$t' = \gamma\left(t - \frac{vx}{c^2}\right) = \frac{5}{3}\left(\frac{5x}{4c} - \frac{4x}{5c}\right)$$

$$t' = \frac{3x}{4c}$$

Earth (x, t)
Initially $(0, 0)$

Finally (x, t')

Spacecraft (x', t')
Initially $(0, 0)$

Finally $(0, t')$

$$d = \frac{x}{\gamma} = \frac{3}{5}x$$

(a) $t' = \frac{3x}{4c}$

(b) $d = \frac{3x}{5}$

Q10. $\frac{dy}{dt} = -0.9c$

$\frac{dy'}{dt'} = \frac{dy}{dt} \frac{dt}{dt'} = \frac{u_y}{\gamma_B} = \frac{-0.9c}{\gamma_B}$

$u_x = 0.9c$

$u_y = -0.9c$

$$t' = \gamma_B\left(t - \frac{v_B x}{c^2}\right) \Rightarrow \frac{dt'}{dt} = \gamma_B\left(1 - \frac{v_B}{c^2} \frac{dx}{dt}\right) = \gamma_B$$

$$x' = \gamma_B(x - v_B t) \Rightarrow \frac{dx'}{dt} = \gamma_B\left(\frac{dx}{dt} - v_B\right) = -v_B \gamma_B$$

$$\frac{dx'}{dt'} = \frac{dx'}{dt} \frac{dt}{dt'} = \frac{-v_B \gamma_B}{\gamma_B} = -v_B = -0.9c$$

$$u_{A|B} = \sqrt{\left(\frac{dx'}{dt'}\right)^2 + \left(\frac{dy'}{dt'}\right)^2} = 0.98c$$

Q11. $(0, t'_1) \rightarrow$ first flash
 $(0, t'_1 + \Delta t'_1) \rightarrow$ second flash.



Note! Correction in Q.

Initial position of $O' =$ same as O .

$$c(t'_1 + \Delta t'_1) = \frac{5}{r} + v(t'_1 + \Delta t'_1) \quad \text{--- (i)}$$

$$ct'_1 + vt'_1 = \frac{5}{r} \quad \text{--- (ii)}$$

$$\Rightarrow \frac{5}{r} + (v - c) \left(\frac{5}{r(v+c)} + \Delta t'_1 \right) = 0$$

$$\Rightarrow 5\sqrt{1-\beta^2} \left(1 + \frac{\beta-1}{\beta+1} \right) + c(\beta-1)\Delta t'_1 = 0$$

$$\Rightarrow \frac{5}{c\Delta t'_1} \left(\frac{2\beta}{\beta+1} \right) = \sqrt{1-\beta}$$

$$\Rightarrow \frac{10}{c\Delta t'_1} = \frac{\sqrt{1-\beta^2}}{\beta} \Rightarrow \frac{100}{c^2(\Delta t'_1)^2} = \frac{1-\beta}{\beta^2}$$

$$\boxed{\beta = \frac{c^2(\Delta t'_1)^2}{100 + c^2(\Delta t'_1)^2} \approx 0.132} \quad \text{Solving}$$

Q12. $\frac{dx'}{dt'} = \frac{d[\gamma(x - vt)]}{d[\gamma(t - \frac{v}{c^2}x)]} = \frac{dx - v\beta dt}{dt - \frac{v\beta}{c^2}dx}$

$\vec{v}$ (rest) $\vec{A}$

$$= \frac{-2v}{1 + \frac{v^2}{c^2}} \quad \gamma' = \frac{1}{\sqrt{1 - \left(\frac{-2v}{1 + \frac{v^2}{c^2}} \right)^2}}$$

$$\frac{1}{\gamma'} = 0.6 \Rightarrow 0.8 = \frac{2\sqrt{c}}{1 + \frac{v^2}{c^2}}$$

$$\therefore \boxed{v = 0.5c} \quad \underline{\text{Sol}^n}$$

$$\beta^2 - 2.5\beta + 1 = 0$$

$$\beta = \frac{2.5 \pm 1.5}{2} = 0.5$$

Lecture 6 (Jan 29)

[No lec on Jan 26]

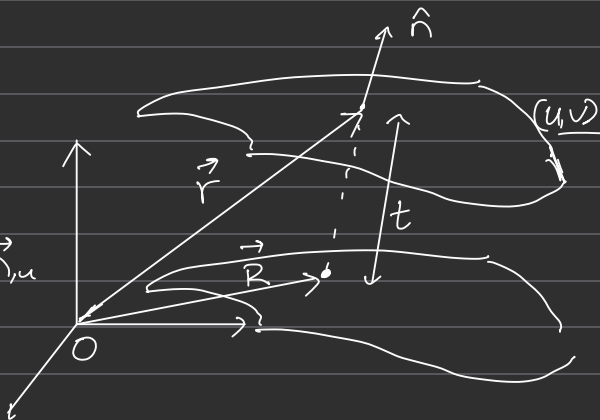
IInd Fundamental Form

$$\vec{R} = \vec{r} - t\vec{n}$$

$$\begin{aligned}\tilde{E} &= \vec{R}_u \cdot \vec{R}_u \\ &= \vec{r}_u \cdot \vec{r}_u - 2t\vec{r}_u \cdot \vec{n}_u + t^2 \vec{n}_u \cdot \vec{n}_u\end{aligned}$$

$$\begin{aligned}\tilde{F} &= \vec{R}_u \cdot \vec{R}_v \\ &= \vec{r}_u \cdot \vec{r}_v - t\vec{r}_u \cdot \vec{n}_v - t\vec{r}_v \cdot \vec{n}_u \\ &\quad + t^2 \vec{n}_u \cdot \vec{n}_v\end{aligned}$$

$$\tilde{G} = \vec{r}_v \cdot \vec{r}_v - 2t\vec{r}_v \cdot \vec{n}_v + t^2 \vec{n}_v \cdot \vec{n}_v$$



$$\therefore \left. \frac{\partial}{\partial t} \left[\tilde{E} du^2 + 2\tilde{F} du dv + \tilde{G} dv^2 \right] \right|_{t=0}$$

$$= \left[\underbrace{(\vec{r}_{uu} \cdot \vec{r})}_{L} du^2 - 2 \underbrace{(\vec{r}_{uv} \cdot \vec{r})}_{M} du dv + \underbrace{(\vec{r}_{vv} \cdot \vec{r})}_{N} dv^2 \right]_{t=0}$$

Gaussian Curvature:

$$= \frac{\det \begin{bmatrix} L & M \\ M & N \end{bmatrix}}{\det \begin{bmatrix} E & F \\ F & G \end{bmatrix}}$$

Proper Derivation

$$\vec{R} = \vec{r} - t\vec{n}$$

$$\begin{aligned} ds^2 &= (\vec{R}_u du + \vec{R}_v dv)^2 \\ &= [(\vec{r}_u - t\vec{n}_u) du + (\vec{r}_v - t\vec{n}_v) dv]^2 \\ &= (\vec{r}_u - t\vec{n}_u)^2 du^2 + 2(\vec{r}_u - t\vec{n}_u)(\vec{r}_v - t\vec{n}_v) du dv \\ &\quad + (\vec{r}_v - t\vec{n}_v)^2 dv^2 \end{aligned}$$

$$\begin{aligned} \frac{d(ds^2)}{dt} \Big|_{t=0} &= \left\{ 2(\vec{r}_u - t\vec{n}_u) \cdot (-\vec{n}_u) du^2 + 2[(\vec{r}_v - t\vec{n}_v) \cdot (-\vec{n}_u) \right. \\ &\quad \left. + (\vec{r}_u - t\vec{n}_u) \cdot (-\vec{n}_v)] du dv + [2(\vec{r}_v - t\vec{n}_v) \cdot (-\vec{n}_v)] dv^2 \right\} \\ &= - \left[2(\vec{r}_u \cdot \vec{n}_u) du^2 + 2(\vec{r}_v \cdot \vec{n}_u + \vec{r}_u \cdot \vec{n}_v) du dv + 2(\vec{r}_v \cdot \vec{n}_v) dv^2 \right] \\ &= \left[2 \left(-\frac{\partial}{\partial u} (\vec{r}_u \cdot \vec{n}) + \vec{r}_{uu} \cdot \vec{n} \right) du^2 + 2 \left(-\frac{\partial}{\partial u} (\vec{r}_v \cdot \vec{n}) + \vec{r}_{uv} \cdot \vec{n} \right) \right. \\ &\quad \left. - \frac{\partial}{\partial v} (\vec{r}_v \cdot \vec{n}) + \vec{r}_{vv} \cdot \vec{n} \right) du dv + 2 \left(-\frac{\partial}{\partial v} (\vec{r}_v \cdot \vec{n}) + \vec{r}_{vv} \cdot \vec{n} \right) dv^2 \right] \\ &= 2 \left[\underbrace{(\vec{r}_{uu} \cdot \vec{n})}_L du^2 + 2 \underbrace{(\vec{r}_{uv} \cdot \vec{n})}_M du dv + \underbrace{(\vec{r}_{vv} \cdot \vec{n})}_N dv^2 \right] \end{aligned}$$

$$\therefore \frac{d(ds^2)}{dt} \Big|_{t=0} = 2(L du^2 + 2M du dv + N dv^2)$$

$$\mathbb{II}(u, v) = L du^2 + 2M du dv + N dv^2$$

Gaussian Curvature :

$$K = \frac{\det \begin{bmatrix} L & M \\ M & N \end{bmatrix}}{\det \begin{bmatrix} E & F \\ F & G \end{bmatrix}}$$

$$\mathbb{II} \quad L = \vec{r}_{uu} \cdot \vec{n}$$

$$M = \vec{r}_{uv} \cdot \vec{n}$$

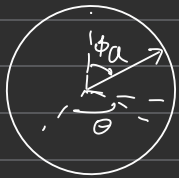
$$N = \vec{r}_{vv} \cdot \vec{n}$$

$$\mathbb{I} \quad E = \vec{r}_u \cdot \vec{r}_u$$

$$F = \vec{r}_u \cdot \vec{r}_v$$

$$G = \vec{r}_v \cdot \vec{r}_v$$

Ex. Sphere



$$\vec{r} = a \hat{n}$$

$$\therefore L = \vec{r}_{\theta\theta} \cdot \vec{n} = \frac{1}{a} \vec{r}_{\theta\theta} \cdot \vec{r}$$

$$M = \vec{r}_{\theta\phi} \cdot \vec{n} = \frac{1}{a} \vec{r}_{\theta\phi} \cdot \vec{r}$$

$$N = \vec{r}_{\phi\phi} \cdot \vec{n} = \frac{1}{a} \vec{r}_{\phi\phi} \cdot \vec{r}$$

$$\vec{r} = a \cos\theta \sin\phi \hat{e}_1 + a \sin\theta \sin\phi \hat{e}_2 + a \cos\phi \hat{e}_3$$

$$\vec{r}_{\theta} = -a \sin\theta \sin\phi \hat{e}_1 + a \cos\theta \sin\phi \hat{e}_2$$

$$\vec{r}_{\phi} = a \cos\theta \cos\phi \hat{e}_1 + a \sin\theta \cos\phi \hat{e}_2 - a \sin\phi \hat{e}_3$$

$$\vec{r}_{\theta\theta} = -a \cos\theta \sin\phi \hat{e}_1 - a \sin\theta \sin\phi \hat{e}_2$$

$$\vec{r}_{\theta\phi} = -a \sin\theta \cos\phi \hat{e}_1 + a \cos\theta \cos\phi \hat{e}_2$$

$$\vec{r}_{\phi\phi} = -a \cos\theta \sin\phi \hat{e}_1 - a \sin\theta \sin\phi \hat{e}_2 - a \cos\phi \hat{e}_3$$

$$\therefore L = \vec{r}_{\theta\theta} \cdot \hat{n} = a \sin^2\phi$$

$$E = a^2 \sin\theta$$

$$M = \vec{r}_{\theta\phi} \cdot \hat{n} = 0$$

$$F = 0$$

$$N = \vec{r}_{\phi\phi} \cdot \hat{n} = -a$$

$$G = a^2$$

$$\therefore L = \frac{1}{a} E, \quad M = 0, \quad N = \frac{1}{a} G.$$

$$\therefore \text{Gaussian Curvature} = \frac{1}{a^2}$$

Curved geometry in higher dimensions

Generalised coordinates : $(q^1, q^2, \dots, q^n)$

$$d\vec{s} = dq^i e_i = dq_j e^j$$

$$ds^2 = e_i \cdot e_j dq^i dq^j$$

Metric : $g_{ij} = \vec{e}_i \cdot \vec{e}_j$

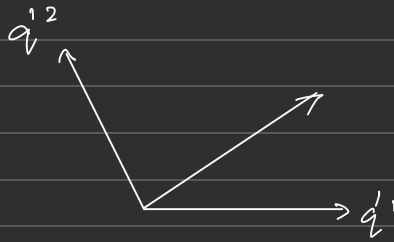
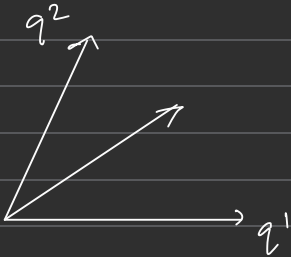
Vectors —

$$\vec{A} = A^i \vec{e}_i$$

$$\vec{e}^j \cdot \vec{A} = \vec{e}^j \cdot A^i \vec{e}_i = A^i \delta_i^j = A^j$$

$$A_i = \vec{e}_i \cdot \vec{A} = \vec{e}_i \cdot \vec{e}_j A^j = g_{ij} A^j$$

Coordinate Transformation



$$dq'^i = \frac{\partial q'^i}{\partial q^j} dq^j$$

$$dq^i = \frac{\partial q^i}{\partial q'^j} dq'^j$$

$$\Rightarrow dq'^i = \underbrace{\frac{\partial q'^i}{\partial q^j}}_{\Lambda^i_{\cdot j}} \underbrace{\frac{\partial q^j}{\partial q'^k}}_{\bar{\Lambda}^j_{\cdot k}} dq'^k$$

$$d\vec{s} = \vec{e}_i dq^i = \vec{e}'_i dq'^i$$

$$\vec{e}^j \cdot (\vec{e}_i dq^i) = \vec{e}^j \cdot (\vec{e}'_i dq'^i)$$

$$dq^j = (\vec{e}^j \cdot \vec{e}'_i) dq'^i$$

$$\therefore \frac{dq^j}{dq'^i} = \vec{e}^j \cdot \vec{e}'_i$$

$$dq^i = \frac{\partial q^i}{\partial q'^j} dq'^j = (\vec{e}^i \cdot \vec{e}'_j) dq'^j$$

HW Show that $\vec{e}'_j = \frac{\partial q^i}{\partial q'^j} \vec{e}_i$

$$\vec{ds} = dq^i \vec{e}_i = dq'^j \vec{e}'_j \Rightarrow dq^i \vec{e}_i = \frac{\partial q^i}{\partial q'^j} dq'^j \vec{e}'_j = dq'^j \vec{e}'_j$$

$$\Rightarrow dq'^j \left[\frac{\partial q^i}{\partial q'^j} \vec{e}_i - \vec{e}'_j \right] = 0 \Rightarrow \vec{e}'_j = \frac{\partial q^i}{\partial q'^j} \vec{e}_i \quad \left(\text{from Linear Independence} \right)$$

Covariant & Contravariant

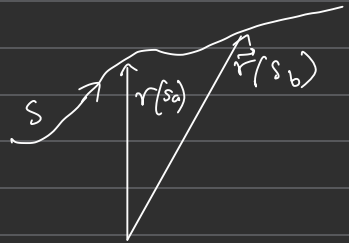
$$\vec{V} = V^i \vec{e}_i = V_j \vec{e}^j$$

V^i transforms like $dq^i \Rightarrow V^i$ is contravariant

$$V^i = \frac{\partial q^i}{\partial q'^j} V^j$$

① tangent vector: $V^i = \frac{dx^i}{ds}$

$$V'^i = \frac{dx'^i}{ds} = \frac{dx'^i}{dx^j} \frac{dx^j}{ds} = \frac{dx'^i}{dx^j} V^j$$



② Scalar field $\phi(x^i)$

$$E_i = \frac{\partial \phi}{\partial x^i}$$

$$E'_i = \frac{\partial \phi}{\partial x'^i} = \frac{\partial \phi}{\partial x^j} \frac{\partial x^j}{\partial x'^i} = E_j \frac{\partial x^j}{\partial x'^i} \rightarrow \underline{\underline{\text{Covariant}}}$$

Covariant & Contravariant

Covariant: Transforms opposite to dq^i

Contravariant: Transforms like dq^i $\left[d\tilde{q}^i = \frac{\partial \tilde{q}^i}{\partial q^j} dq^j \right]$

★ Scalar:

$$f_i = \frac{\partial \phi}{\partial x^i} \Rightarrow \tilde{f}_i = \frac{\partial \phi}{\partial \tilde{x}^i} = \frac{\partial x^j}{\partial \tilde{x}^i} \frac{\partial \phi}{\partial x^j} = \frac{\partial x^j}{\partial \tilde{x}^i} f_j \quad [\text{Covariant}]$$

★ Basis Vectors:

$$e_i = \frac{\partial \vec{r}}{\partial q^i} \Rightarrow \tilde{e}_i = \frac{\partial \vec{r}}{\partial \tilde{q}^i} = \frac{\partial q^j}{\partial \tilde{q}^i} \frac{\partial \vec{r}}{\partial q^j} = \frac{\partial q^j}{\partial \tilde{q}^i} e_j \quad [\text{Covariant}]$$

$e^i \Rightarrow$ Contravariant

$$e_i \cdot e^i = \delta_i^i \Rightarrow \frac{\partial \tilde{q}^j}{\partial q^i} \tilde{e}_j \cdot e^i = \delta_i^i = \delta_j^j = \tilde{e}_j \cdot \tilde{e}^j$$

$$\Rightarrow \tilde{e}_j \cdot \left[\frac{\partial \tilde{q}^j}{\partial q^i} e^i - \tilde{e}^j \right] = 0 \xrightarrow{\text{Arbitrary}} \tilde{e}^j = \frac{\partial \tilde{q}^j}{\partial q^i} e^i$$

★ $dq_i \Rightarrow$ Covariant

$$d\vec{r} = d\tilde{q}_i \tilde{e}^i = dq_j e^j = dq_j \frac{\partial \tilde{q}^j}{\partial q^i} \tilde{e}^i$$

$$\text{By linear Independence} \rightarrow d\tilde{q}_i = \frac{\partial \tilde{q}^j}{\partial q^i} dq_j$$

★ Components of a vector $\vec{v} = v_i e^i = v^i e_i$

$$v_i \Rightarrow \text{Covariant} \quad v_i e^i = v_i \frac{\partial \tilde{q}^j}{\partial q^i} \tilde{e}^j = \tilde{v}_j \tilde{e}^j \xrightarrow{\text{L.I.}} \tilde{v}_j = \frac{\partial \tilde{q}^j}{\partial q^i} v_i$$

$$v^i \Rightarrow \text{Contravariant} \quad v^i e_i = v^i \frac{\partial \tilde{q}^j}{\partial q^i} \tilde{e}_j = \tilde{v}^j \tilde{e}_j \xrightarrow{\text{L.I.}} \tilde{v}^j = \frac{\partial \tilde{q}^j}{\partial q^i} v^i$$

Tensor

$$T \underset{\sim}{=} T^{ij} e_i \otimes e_j = T^{ij} \vec{e}^k g_{ki} \otimes \vec{e}^l g_{lj} = (T^{ij} g_{ki} g_{lj}) \vec{e}^k \otimes \vec{e}^l$$

$$\therefore \underline{T_{ij} = g_{ki} g_{lj} T^{kl}}$$

Lecture 7 (2 Feb)

Rank k tensor

Contravariant -

$$g'^{i_1 i_2 \dots i_k} = \frac{\partial x^{i_1}}{\partial x'^{j_1}} \frac{\partial x^{i_2}}{\partial x'^{j_2}} \dots \frac{\partial x^{i_k}}{\partial x'^{j_k}} g^{j_1 j_2 \dots j_k}$$

Covariant -

$$g'_{i_1 i_2 \dots i_k} = \frac{\partial x^{j_1}}{\partial x'^{i_1}} \frac{\partial x^{j_2}}{\partial x'^{i_2}} \dots \frac{\partial x^{j_k}}{\partial x'^{i_k}} g_{j_1 j_2 \dots j_k}$$

Computing derivatives in curvilinear coordinates:

Plane polar coordinates -

$$\vec{e}_r = \cos\varphi \vec{i} + \sin\varphi \vec{j}$$

$$\vec{e}_\varphi = -r \sin\varphi \vec{i} + r \cos\varphi \vec{j}$$

$$\frac{\partial \vec{e}_r}{\partial r} = 0, \quad \frac{\partial \vec{e}_r}{\partial \varphi} = \frac{\vec{e}_\varphi}{r}, \quad \frac{\partial \vec{e}_\varphi}{\partial r} = \frac{\vec{e}_\varphi}{r}, \quad \frac{\partial \vec{e}_\varphi}{\partial \varphi} = -r \vec{e}_r$$

$$\frac{\partial \vec{e}_i}{\partial x^j} = \Gamma_{ij}^k \vec{e}_k$$

$$\vec{A} = A^i \vec{e}_i$$

$$d\vec{A} = dA^i \vec{e}_i + A^i d\vec{e}_i = dA^i \vec{e}_i + A^i \Gamma_{ij}^k \vec{e}_k dx^j$$

$$\therefore \frac{\partial \vec{A}}{\partial x^j} = \frac{\partial A^i}{\partial x^j} \vec{e}_i + A^i \Gamma_{ij}^k \vec{e}_k$$

$$\Rightarrow \boxed{\frac{\partial \vec{A}}{\partial x^j} = \left(\frac{\partial A^k}{\partial x^j} + A^i \Gamma_{ij}^k \right) \vec{e}_k}$$

$$d\vec{A} = \underbrace{(\nabla_j A^k)}_{\text{covariant derivative}} \vec{e}_k dx^j$$

$$\nabla_j A^k = \partial_j A^k + A^i \Gamma_{ij}^k$$

line element -

$$d\vec{s} = \vec{e}_i dx^i = \frac{\partial \vec{s}}{\partial x^i} dx^i \quad \therefore \vec{e}_i = \frac{\partial \vec{s}}{\partial x^i}$$

$$\frac{\partial \vec{e}_i}{\partial x^j} = \frac{\partial^2 \vec{s}}{\partial x^i \partial x^j} = \frac{\partial^2 \vec{s}}{\partial x^j \partial x^i} = \frac{\partial \vec{e}_j}{\partial x^i}$$

$$\therefore \boxed{\Gamma_{ij}^k = \Gamma_{ji}^k}$$

Metric -

$$g_{ij} = \vec{e}_i \cdot \vec{e}_j$$

$$\begin{aligned} \Rightarrow dg_{ij} &= d\vec{e}_i \cdot \vec{e}_j + \vec{e}_i \cdot d\vec{e}_j \\ &= \left(\frac{\partial \vec{e}_i}{\partial x^k} \cdot \vec{e}_j + \vec{e}_i \cdot \frac{\partial \vec{e}_j}{\partial x^k} \right) dx^k \end{aligned}$$

$$\downarrow = \left(\Gamma_{ik}^m \vec{e}_m \cdot \vec{e}_j + \vec{e}_i \cdot \Gamma_{jk}^m \vec{e}_m \right) dx^k$$

$$\frac{\partial g_{ij}}{\partial x^k} = \left(\Gamma_{ik}^m g_{mj} + \Gamma_{jk}^m g_{mi} \right) dx^k$$

$$\Rightarrow \frac{\partial g_{ij}}{\partial x^k} = \Gamma_{ik}^m g_{mj} + \Gamma_{jk}^m g_{mi}$$

$$\Rightarrow \frac{\partial g_{ij}}{\partial x^k} = \Gamma_{jk}^k g_{ik} + \Gamma_{ik}^k g_{kj}$$

$$\Rightarrow \frac{\partial g_{ik}}{\partial x^j} = \Gamma_{lj}^k g_{ik} + \Gamma_{ij}^k g_{kl}$$

Now:

$$\frac{\partial g_{il}}{\partial x_i} + \frac{\partial g_{lj}}{\partial x_i} - \frac{\partial g_{ij}}{\partial x^l} = 2 \underbrace{\Gamma_{ij}^k g_{lk}}_{\text{Christoffel Symbols}}$$

Note:

Christoffel Symbols are not Tensors.

$$\tilde{\Gamma}_{jk}^i = \frac{\partial \tilde{g}_j}{\partial \tilde{q}^k} \cdot \tilde{g}^i = \frac{\partial}{\partial \tilde{q}^k} \left[\frac{\partial q^a}{\partial \tilde{q}^i} g_a \right] \cdot \left(\frac{\partial \tilde{q}^i}{\partial q^b} g^b \right)$$

$$= \underbrace{\left(\frac{\partial^2 q^b}{\partial \tilde{q}^i \partial \tilde{q}^k} \right)}_{\neq 0} \frac{\partial \tilde{q}^i}{\partial q^b} + \frac{\partial \tilde{q}^i}{\partial q^b} \frac{\partial q^a}{\partial \tilde{q}^j} \frac{\partial q^r}{\partial \tilde{q}^k} \Gamma_{qr}^b$$

Lecture 8

Covariant Derivative

$$\nabla_j A^i = \partial_j A^i + \Gamma_{jk}^i A^k$$

$$de_i = \Gamma_{ij}^k \vec{e}_k dx^j$$

$$\nabla_k g_{ij} = 0$$

tensor of rank 3

HW

$$\nabla_j A_k = \partial_j A_k - \Gamma_{jk}^i A_i$$

$$de^i = -\Gamma_{jk}^i \vec{e}^j dx^k$$

Covariant derivative & parallel transport

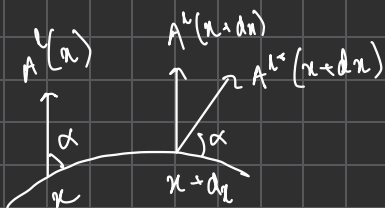
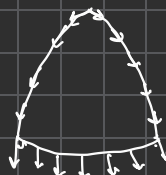
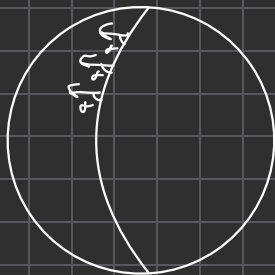
Euclidean Space:



→ geodesic



Curved geometry:



$$\begin{aligned} & A^i(x+dx) - A^i(x) \\ &= A^i(x) + \frac{\partial A^i}{\partial x^j} dx^j - (A^i(x) + \delta A^i) \end{aligned}$$

HW

$$\delta A^i = -\Gamma_{jk}^i dx^j A^k$$

$$= \frac{\partial A^i}{\partial x^j} dx^j + \Gamma_{jk}^i dx^j A^k$$

$$= \nabla_g A^l dx^g$$

geodesic eqⁿ

$$\ddot{x}(x(\lambda)) = \ddot{x}(\lambda) \vec{e}_i$$

A vector field $\vec{v} = v^i \vec{e}_i$

Flat: $\frac{dv^i}{d\lambda} = 0$

Curved space

$$\begin{aligned} \frac{D\vec{v}}{D\lambda} &= \frac{dv^i}{d\lambda} \vec{e}_i + v^i \frac{d\vec{e}_i}{d\lambda} \\ &= \frac{dv^i}{d\lambda} \vec{e}_i + v^i \frac{dx^j}{d\lambda} \frac{d\vec{e}_i}{dx^j} \\ &= 0 + v^i \frac{dx^j}{d\lambda} \Gamma_{ij}^k \vec{e}_k \end{aligned}$$

Special case: $\vec{v} = \text{velocity}$

$$\frac{dx^i}{d\lambda} = v^i$$

$$\frac{D\vec{v}}{d\lambda} = \frac{dv^k}{d\lambda} \vec{e}_k + \Gamma_{ij}^k v^i v^j \vec{e}_k = 0$$

$\vec{x}$ Particle 1

$$\vec{a} = -\vec{\nabla}\phi$$

EOM

$$\frac{d^2 x^i}{dt^2} = -g^{ij} \frac{\partial \phi(x^k)}{\partial x^j}$$

$\vec{x} + \vec{\eta}$ particle 2

~~$$\frac{d^2 (x^i + \eta^i)}{dt^2} = -g^{ij} \frac{\partial \phi(x^k + \eta^k)}{\partial x^j} = -g^{ij} \left(\frac{\partial \phi}{\partial x^j} + \eta^l \frac{\partial^2 \phi}{\partial x^l \partial x^j} \right)$$~~

$$\frac{d^2 (x^i + \eta^i)}{dt^2} = \frac{\partial \phi(x^k + \eta^k)}{\partial x^i} = \frac{\partial \phi(x^k)}{\partial x^i} + \frac{\partial^2 \phi}{\partial x^i \partial x^j} \Big|_{x^k} \eta^j$$

$$\Rightarrow \frac{d^2 \eta^i}{dt^2} = -g^{ij} \eta_j^l \frac{\partial^2 \phi}{\partial x^l \partial x^i}$$

$$\frac{d^2 \vec{\eta}}{dt^2} = \nabla(\nabla\phi) \cdot \vec{\eta} \quad \hookrightarrow \text{(Force)}$$

$$= (\vec{\eta} \cdot \vec{\nabla} \underline{F^i}) \quad \text{tidal force}$$

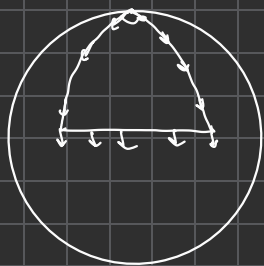
$$\frac{\partial^2 \phi}{\partial x^i \partial x^i} \rightarrow \text{is this a tensor?} \rightarrow \text{No}$$

Proof:

$$\frac{\partial \phi}{\partial x^i} = v_i \rightarrow \text{co-vector}$$

$$v_i' = \frac{\partial x^j}{\partial x'^i} v_j$$

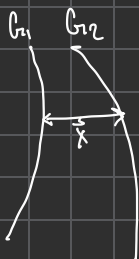
$$\frac{\partial v_i'}{\partial x'^k} = \frac{\partial}{\partial x'^k} \left(\frac{\partial x^j}{\partial x'^i} v_j \right) = \frac{\partial^2 x^j}{\partial x'^k \partial x'^i} v_j + \frac{\partial x^j}{\partial x'^i} \frac{\partial v_j}{\partial x'^k}$$



$$d\vec{A} = \nabla_j A^i \vec{e}_i dx^j$$

$$\frac{d\vec{A}}{d\lambda} = \frac{dx^j}{d\lambda} \nabla_j \vec{A} \rightarrow \equiv U^j$$

$$\nabla_\lambda \vec{A} \equiv \frac{d\vec{A}}{d\lambda} = U^j \nabla_j \vec{A}$$

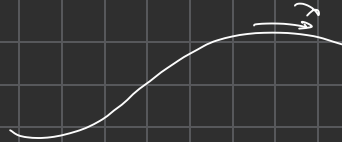


$$\begin{aligned} \vec{\nabla} &\equiv \nabla_x \vec{x} = v^j \nabla_j \vec{x} \\ &= \left(\partial_j x^i + \Gamma_{kj}^i x^k \right) \frac{dx^j}{d\lambda} \vec{e}_i \\ &= \underbrace{\left(\frac{dx^i}{d\lambda} + \Gamma_{kj}^i v^j x^k \right)}_{v^i} \vec{e}_i \end{aligned}$$

$$\nabla_{\vec{v}} \vec{v} = \nabla_{\vec{v}} (\nabla_{\vec{v}} \vec{x})$$

$$= v^j \nabla_j \vec{v}$$

Lecture 9 (9 Feb)



$$\frac{d\vec{A}}{d\lambda} = \frac{dx^j}{d\lambda} \nabla_j \vec{A}$$

$$\nabla_u \vec{A} = \frac{d\vec{A}}{d\lambda} = u^j \nabla_j \vec{A} \quad u^j = \frac{dx^j}{d\lambda}$$

$$\vec{v} = \nabla_u \vec{x} \quad \vec{v}: \text{Velocity.}$$

$$v^i = \frac{dx^i}{d\lambda} + \Gamma_{kj}^i u^j x^k$$

Acceleration of the separation vector —

$$\vec{\omega} = \nabla_u (\nabla_u \vec{x})$$

$$\omega^i = u^j \nabla_j v^i = u^j (\partial_j v^i + \Gamma_{jm}^i v^m)$$

$$= \frac{dx^i}{d\lambda} \partial_j \left(\frac{dx^i}{d\lambda} + \Gamma_{lm}^i x^l u^m \right) + \Gamma_{jm}^i u^j \left(\frac{dx^m}{d\lambda} + \Gamma_{ab}^m x^a u^b \right)$$

$$= \frac{d^2 x^i}{d\lambda^2} + \left(\Gamma_{lm}^i x^l \frac{du^m}{d\lambda} + 2 \Gamma_{lj}^i u^j \frac{dx^l}{d\lambda} + x^l u^m u^j \frac{\partial \Gamma_{lm}^i}{\partial x^j} \right) + \Gamma_{jm}^i \Gamma_{ab}^m x^a u^b u^j$$

$$\nabla_j \vec{A} = (\partial_j A^i) e_i + A^k \Gamma_{jk}^i e_i = (\partial_j A^i + \Gamma_{jk}^i A^k) e_i$$

$$\therefore \vec{v} = \frac{d\vec{x}}{d\lambda}$$

↗ small $(x \neq X \text{ are different})$
 $\hookrightarrow \approx \delta x$

$x^i + X^i$ obeys geodesic equation.

$$\frac{d^2(x^i + X^i)}{d\lambda^2} + \Gamma_{jk}^i \frac{d(x^j + X^j)}{d\lambda} \frac{d(x^k + X^k)}{d\lambda} = 0$$

$$\Rightarrow \frac{d^2 x^i}{d\lambda^2} + \frac{d^2 X^i}{d\lambda^2} + \left(\Gamma_{jk}^i(x) + \frac{\partial \Gamma_{jk}^i}{\partial x^l} X^l \right) \left(\frac{dx^j}{d\lambda} + \frac{dX^j}{d\lambda} \right) \left(\frac{dx^k}{d\lambda} + \frac{dX^k}{d\lambda} \right) = 0$$

$\underbrace{\quad}_{u^j} \quad \underbrace{\quad}_{u^k}$

something, something - -

$$\omega^i = - \left[\frac{\partial \Gamma_{jk}^i}{\partial x^l} X^l u^j u^k - \frac{\partial \Gamma_{lm}^i}{\partial x^j} u^j u^m X^l + \Gamma_{kj}^m u^k u^j \Gamma_{lm}^i X^l - \Gamma_{jm}^i \Gamma_{lk}^m X^l u^k u^j \right]$$

Note: $\frac{du^m}{d\lambda} = \frac{d}{d\lambda} \left(\frac{dx^m}{d\lambda} \right) = \frac{d^2 x^m}{d\lambda^2} \xrightarrow{\text{Geodesic Eq}} -\Gamma_{ij}^m u^i u^j$
 (Eq. 6.2)

$$\omega^i = - \left[\frac{\partial \Gamma_{jk}^i}{\partial x^l} - \frac{\partial \Gamma_{lk}^i}{\partial x^j} + \Gamma_{kj}^m \Gamma_{lm}^i - \Gamma_{jm}^i \Gamma_{lk}^m \right] X^l u^j u^k$$

$$\boxed{\nabla_u \nabla_u X^i = -R_{jkl}^i X^l u^j u^k}$$

Geodesic Deviation Equation.
in Riemannian Space.

R_{jkl}^i : Riemann Tensor

HW

$$\textcircled{1} R_{ijkl} = g_{ia} R_{jkl}^a = \frac{1}{2} \left(\frac{\partial^2 g_{il}}{\partial x^j \partial x^k} + \frac{\partial^2 g_{jk}}{\partial x^i \partial x^l} - \frac{\partial^2 g_{ik}}{\partial x^j \partial x^l} - \frac{\partial^2 g_{jl}}{\partial x^i \partial x^k} \right) + g_{mn} (\Gamma_{jk}^m \Gamma_{il}^n - \Gamma_{jl}^m \Gamma_{ik}^n)$$

$$\textcircled{2} R_{ijkl} = -R_{jikl} = -R_{ijlk} = R_{klij}$$

③ Bianchi Identity

$$R_{ijkl} + R_{iklj} + R_{iljk} = 0$$

components in R_{ijkl} in D-dimensions.

$$D_C2 = 1$$

$$\frac{D_C2 (D_C2 - 1)}{2}$$

$$D=2 \Rightarrow \begin{array}{ll} 1111 & 2111 \\ 1112 & 2112 \\ 1121 & 2121 \\ 1122 & 2122 \\ 1211 & 2211 \\ 1212 & 2212 \\ 1221 & 2221 \\ 1222 & 2222 \end{array}$$

$$\begin{array}{l} 1111 \\ 1112 \\ 1122 \\ 1212 \\ 1222 \\ 2222 \end{array} \left. \vphantom{\begin{array}{l} 1111 \\ 1112 \\ 1122 \\ 1212 \\ 1222 \\ 2222 \end{array}} \right\} 6 \text{ components}$$

↳ Antisymmetric Property.

∴ Only 1 component.

$$\underline{R_{1212} \neq 0}$$

$$Q. \quad ds^2 = g_{11}(dx^1)^2 + g_{22}(dx^2)^2$$

$$\Gamma_{11}^1 = \frac{\partial g_1}{\partial x^1} g_1 = \frac{\partial g_1}{\partial x^1} g_1$$

$$\Gamma_{11}^2 = \frac{\partial g_1}{\partial x^1} g_2$$

$$\Gamma_{12}^1 = \frac{\partial g_1}{\partial x^2} g_1$$

$$\Gamma_{12}^2 = \frac{\partial g_1}{\partial x^2} g_2$$

$$\Gamma_{22}^1 = \frac{\partial g_2}{\partial x^2} g_1$$

$$\Gamma_{22}^2 = \frac{\partial g_2}{\partial x^2} g_2$$

$$\Gamma_{ij}^k = \frac{\partial g_i}{\partial x^j} g$$

HW Compute R_{212} .

Geodesic Deviation Equation (Rederivation)

Let a geodesic in some n -dimensional space be represented by the parameter 's' & 't' represents a parameter along the geodesic.

$\therefore x^i(s, t)$ denote coordinates of a point along a geodesic.

Let $X^i = \frac{\partial x^i}{\partial s}(s, t)$: Separation vector b/w 2 close geodesics.

Let $\vec{T} = \frac{\partial \vec{x}}{\partial t}(s, t)$: Tangent vector along the geodesic.

Note:

for any vector $\vec{A}$,

$$\frac{d\vec{A}}{d\lambda} = (\nabla \vec{A}) \cdot \frac{d\vec{x}}{d\lambda} \quad (\nabla \vec{A} : \text{Covariant Gradient})$$

$$\begin{aligned} \text{Proof: } \frac{d\vec{A}}{d\lambda} &= \frac{d(A^i e_i)}{d\lambda} = \frac{dA^i}{d\lambda} e_i + A^i \frac{de_i}{d\lambda} = \frac{dA^i}{d\lambda} e_i + A^i \frac{\partial e_i}{\partial x^j} \frac{dx^j}{d\lambda} \\ &= \frac{\partial A^k}{\partial x^i} \frac{dx^i}{d\lambda} e_k + A^i \Gamma_{ij}^k e_k \frac{dx^j}{d\lambda} = \underbrace{\left(\frac{\partial A^k}{\partial x^i} + A^i \Gamma_{ij}^k \right) e_k}_{\nabla \vec{A} \text{ (Covariant Gradient)}} \underbrace{\frac{dx^j}{d\lambda}}_{\frac{d\vec{x}}{d\lambda}} \end{aligned}$$

$\therefore$ Directional derivative along the tangent $\vec{T}(\lambda) = \frac{d\vec{x}}{d\lambda}$ can be expressed like that.

Define relative Acceleration $\vec{A}$ as the acceleration of Separation vector as we move along the geodesic.

$$\therefore \boxed{\vec{A} = \frac{\partial^2 \vec{x}}{\partial t^2} = \vec{T} \cdot \nabla (\vec{T} \cdot \nabla \vec{x})}$$

The point $(\vec{x} + \vec{X})$ lie over a geodesic. Hence it must satisfy the geodesic equation.

$$\therefore \frac{d^2(x^i + X^i)}{d\tau^2} + \Gamma_{jk}^i(\vec{x} + \vec{X}) \frac{d(x^j + X^j)}{d\tau} \frac{d(x^k + X^k)}{d\tau} = 0$$

Since $\vec{X}$ is small, $\therefore \Gamma_{jk}^i(\vec{x} + \vec{X}) = \Gamma_{jk}^i(\vec{x}) + \frac{\partial \Gamma_{jk}^i}{\partial x^\ell} X^\ell$

Substituting this approximating & neglecting $O(X^2)$ terms, we get:

$$\left(\frac{d^2 x^i}{d\tau^2} + \Gamma_{jk}^i \frac{dx^j}{d\tau} \frac{dx^k}{d\tau} \right) + \frac{d^2 X^i}{d\tau^2} + \Gamma_{jk}^i \frac{dx^j}{d\tau} \frac{dX^k}{d\tau} + \Gamma_{jk}^i \frac{dX^j}{d\tau} \frac{dx^k}{d\tau} + \frac{\partial \Gamma_{jk}^i}{\partial x^\ell} X^\ell \frac{dx^j}{d\tau} \frac{dx^k}{d\tau} + O(X^2) = 0$$

Since $\vec{x}$ also lies over a geodesic, $\frac{d^2 x^i}{d\tau^2} + \Gamma_{jk}^i \frac{dx^j}{d\tau} \frac{dx^k}{d\tau} = 0$

$$\therefore \frac{d^2 X^i}{d\tau^2} + 2 \Gamma_{jk}^i \frac{dx^j}{d\tau} \frac{dX^k}{d\tau} + \frac{\partial \Gamma_{jk}^i}{\partial x^\ell} X^\ell \frac{dx^j}{d\tau} \frac{dx^k}{d\tau} = 0$$

Simplifying our Acceleration vector:

$$\begin{aligned} \vec{A} &= \frac{\partial^2 \vec{X}}{\partial \tau^2} = \frac{\partial}{\partial \tau} \left[\left(\frac{\partial X^i}{\partial \tau} + \frac{dx^j}{d\tau} X^k \Gamma_{jk}^i \right) e_i \right] \\ &= \left(\frac{\partial^2 X^i}{\partial \tau^2} + \frac{d^2 x^j}{d\tau^2} X^k \Gamma_{jk}^i + \frac{dx^j}{d\tau} \frac{\partial X^k}{\partial \tau} \Gamma_{jk}^i + \frac{dx^j}{d\tau} X^k \frac{\partial \Gamma_{jk}^i}{\partial x^\ell} \frac{dx^\ell}{d\tau} \right) e_i \\ &\quad + \underbrace{\left(\frac{\partial X^i}{\partial \tau} + \frac{dx^j}{d\tau} X^k \Gamma_{jk}^i \right) \left(\Gamma_{im}^n e_n \frac{dx^m}{d\tau} \right)}_{\text{Relabeling terms}} \\ &= \left(\Gamma_{jk}^i \frac{dx^j}{d\tau} \frac{dX^k}{d\tau} + \Gamma_{jm}^i \Gamma_{kl}^m \frac{dx^j}{d\tau} \frac{dx^\ell}{d\tau} X^k \right) e_i \quad [\text{Relabeling terms}] \end{aligned}$$

Substituting $\frac{d^2 x^i}{d\tau^2} = -\Gamma_{mn}^i \frac{dx^m}{d\tau} \frac{dx^n}{d\tau}$ (Geodesic Eqⁿ)

$$A^i = \frac{d^2 X^i}{d\tau^2} - \Gamma_{jk}^i \Gamma_{mn}^j \frac{dx^m}{d\tau} \frac{dx^n}{d\tau} X^k + \Gamma_{jk}^i \frac{dx^j}{d\tau} \frac{dX^k}{d\tau} + \frac{\partial \Gamma_{jk}^i}{\partial x^l} \frac{dx^j}{d\tau} \frac{dx^l}{d\tau} X^k \\ + \Gamma_{jk}^i \frac{dx^j}{d\tau} \frac{dX^k}{d\tau} + \Gamma_{jm}^i \Gamma_{kl}^m \frac{dx^j}{d\tau} \frac{dx^l}{d\tau} X^k$$

Substituting: $\frac{d^2 X^i}{d\tau^2} + 2 \Gamma_{jk}^i \frac{dx^j}{d\tau} \frac{dX^k}{d\tau} = - \frac{\partial \Gamma_{jk}^i}{\partial x^l} X^l \frac{dx^j}{d\tau} \frac{dx^k}{d\tau}$

$$A^i = \frac{\partial \Gamma_{jk}^i}{\partial x^l} \frac{dx^j}{d\tau} \frac{dx^l}{d\tau} X^k - \frac{\partial \Gamma_{jk}^i}{\partial x^l} \frac{dx^j}{d\tau} \frac{dx^k}{d\tau} X^l + \Gamma_{jm}^i \Gamma_{kl}^m \frac{dx^j}{d\tau} \frac{dx^l}{d\tau} X^k \\ - \Gamma_{jk}^i \Gamma_{mn}^j \frac{dx^m}{d\tau} \frac{dx^n}{d\tau} X^k$$

$$\Rightarrow A^i = \left(\frac{\partial \Gamma_{jk}^i}{\partial x^l} - \frac{\partial \Gamma_{jl}^i}{\partial x^k} + \Gamma_{lm}^i \Gamma_{jk}^m - \Gamma_{km}^i \Gamma_{jl}^m \right) \frac{dx^j}{d\tau} X^k \frac{dx^l}{d\tau}$$

$$R_{jkl}^i = \frac{\partial \Gamma_{jl}^i}{\partial x^k} - \frac{\partial \Gamma_{jk}^i}{\partial x^l} + \Gamma_{km}^i \Gamma_{jl}^m - \Gamma_{lm}^i \Gamma_{jk}^m \quad \text{Riemannian Tensor}$$

Substituting:

$$\frac{d^2 X^i}{d\tau^2} = -R_{jkl}^i \frac{dx^j}{d\tau} \frac{dx^k}{d\tau} X^l$$

Geodesic Deviation Equation.

Properties of Riemann Tensor:

$$\textcircled{1} R_{ijkl} = g_{ia} R^a_{jkl} = \frac{1}{2} \left(\frac{\partial^2 g_{il}}{\partial x^j \partial x^k} + \frac{\partial^2 g_{jk}}{\partial x^i \partial x^l} - \frac{\partial^2 g_{ik}}{\partial x^j \partial x^l} - \frac{\partial^2 g_{jl}}{\partial x^i \partial x^k} \right) + g_{mn} (\Gamma^m_{jk} \Gamma^n_{il} - \Gamma^m_{jl} \Gamma^n_{ik})$$

$$\textcircled{2} R_{ijkl} = -R_{jikl} = -R_{ijlk} = R_{klij}$$

$$\textcircled{3} \text{ Bianchi Identity: } R_{ijkl} + R_{iklj} + R_{iljk} = 0$$

$$\text{Ricci Tensor: } R_{ij} = R^l_{ilj} \quad (\text{Symmetric})$$

$$\text{Ricci Scalar: } R = R^i_i = g^{ij} R_{ij} = g^{ij} R^l_{ilj}$$

Ricci Tensor:

$$R_{ij} = R^k{}_{ikj}$$

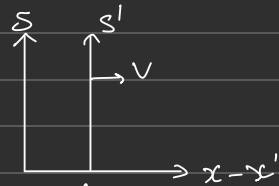
Ricci Scalar

$$R = R^i{}_i = g^{ij} R_{ij} = g^{ij} R^k{}_{ikj}$$

Lecture 10

$$x' = \gamma(x - \beta ct)$$

$$ct' = \gamma(ct - \beta x)$$



$$c^2 \Delta t^2 - \Delta x^2 = c^2 (\Delta t')^2 - (\Delta x')^2 \Rightarrow \text{Invariant.}$$

$$ds^2 = c(dt)^2 - (dx)^2$$

Proper Time: Δt measured by a single clock.

$$\hookrightarrow \Delta x = 0 \Rightarrow \Delta s = c \Delta t$$

$\hookrightarrow$ Proper Invariant.

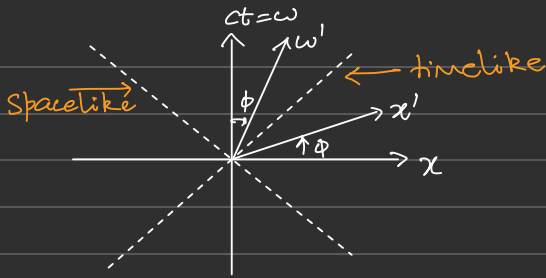
$$\text{Light} \Rightarrow dx = c dt \Rightarrow ds = 0 \quad (\text{Null})$$

$$ds^2 < 0 \Rightarrow c^2 dt^2 > dx^2$$

$$c^2 > v^2 \quad : \text{timelike}$$

$$ds^2 > 0 \Rightarrow c^2 < v^2 \quad : \text{space like.}$$

$$\omega = \beta x$$



$$x' = \gamma(x - \beta \omega)$$

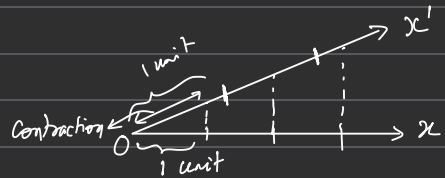
$$\omega' = \gamma(\omega - \beta x)$$

$$x' = 0 \Rightarrow x = \beta \omega \rightarrow \text{Slope} = \frac{1}{\beta}$$

$$\omega' = 0 \Rightarrow \omega = \beta x \rightarrow \text{Slope} = \beta$$

$$\frac{1}{\beta} \times \beta = 1$$

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{bmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$



$$\Rightarrow X'^{\alpha} = \Lambda^{\alpha}_{\beta} X^{\beta}$$

$$dx'^{\mu} = \Lambda^{\mu}_{\nu} dx^{\nu}$$

Note:
 $\uparrow$
 Time space
 Greek Indices: 0, 1, 2, 3
 Latin Indices: 1, 2, 3

$$\underline{\text{Hw}} \quad \det \Lambda = 1$$

(Pure Shear Transformation)

$$dx^{\mu} dx_{\mu} = c^2 dt^2 - dx^2$$

$$dx^{\mu} = (cdt, d\vec{x})$$

$$dx_{\mu} = (cdt, -d\vec{x})$$

$$\underline{\text{4-Velocity}}: u^{\mu} = \frac{dx^{\mu}}{d\tau} = \left(c \frac{dt}{d\tau}, \frac{d\vec{x}}{dt} \frac{dt}{d\tau} \right) = \gamma(c, u^i)$$

$$\text{Hw } u_{\mu} = \gamma(c, -u^i)$$

$$u^{\mu} u_{\mu} = \gamma^2 (c^2 - u^2) = \underline{\underline{c^2}}$$

4-Momentum:

$$p^\alpha = m u^\alpha = m \gamma(c, u^i)$$

$$p_\alpha = m u_\alpha = m \gamma(c, u^i)$$

$$p^\alpha p_\alpha = m^2 c^2 = mE$$

4-Acceleration

$$a^\alpha = \frac{du^\alpha}{d\tau}$$

4-Force

$$F^\alpha = \frac{dp^\alpha}{d\tau} = \frac{d}{d\tau} \left(\frac{E}{c}, p^i \right) = \left(\frac{1}{c} \frac{dt}{d\tau} \frac{dE}{dt}, \frac{dp^i}{d\tau} \right)$$

$$\vec{F} = \gamma \left(\frac{\vec{F} \cdot \vec{u}}{c}, \vec{F} \right)$$

Equation of Motion in 4D

$$F^\alpha = \frac{dp^\alpha}{d\tau}$$

Electromagnetic Force: $\frac{dp^\alpha}{d\tau} = q F^{\alpha\beta} u_\beta$ (H.W.)

(Force $E\vec{v}$)

→ Field Tensor.

(Antisymmetric 4×4 Tensor)

$$F^{\alpha\beta} = \begin{bmatrix} 0 & -\frac{E_x}{c} & -\frac{E_y}{c} & -\frac{E_z}{c} \\ \frac{E_x}{c} & 0 & -B_z & B_y \\ \frac{E_y}{c} & B_z & 0 & -B_x \\ \frac{E_z}{c} & -B_y & B_x & 0 \end{bmatrix}$$

(E_x, E_y, E_z) : Electric Field

(B_x, B_y, B_z) : Magnetic Field.

Lagrangian for a pde in 4D

$$L = \frac{1}{2} m \left(\frac{ds}{d\tau} \right)^2 = \frac{1}{2} m g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = \frac{1}{2} m g_{\mu\nu} u^\mu u^\nu$$

$(ds^2 = g_{\mu\nu} dx^\mu dx^\nu)$

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\sigma}^\mu u^\nu u^\sigma = 0 \quad \text{Geodesic Eqn}$$

Lecture 11 (19 Feb)

Uniform Acceleration in STR

$$\vec{F} = m\vec{a} \quad (\text{constant})$$

Frame: $S \quad S' \xrightarrow{x-x'}$

$u_x = 0 \quad u_x = \frac{u'_x - v}{1 - \frac{u'_x v}{c^2}} \quad u'_x = v \rightarrow \text{changes } \therefore$
 $S \text{ is uniformly accelerated.}$
 $S' \text{ is inertial.}$

4-velocity -

$$U^0 = \gamma(U'^0 - \beta U'^x) \quad U^\mu = \gamma(c, \vec{u})$$

$$U^x = \gamma(U'^x - \beta U'^0) \quad U'^\mu = \gamma(c, \vec{u}')$$

$$U^0 = c\gamma^2(1 - \beta^2) = c$$

$$u_x = \frac{u'_x - v}{1 - \frac{u'_x v}{c^2}} \Rightarrow du_x = \frac{(du'_x - dv)\left(1 - \frac{u'_x v}{c^2}\right) - (u'_x - v)\left(-\frac{du'_x v}{c^2} - \frac{u'_x dv}{c^2}\right)}{\left(1 - \frac{u'_x v}{c^2}\right)^2}$$

$$= \frac{\left[1 - \frac{u'_x v}{c^2} + (u'_x - v)\frac{v}{c^2}\right]du'_x + \left[-(1 - \frac{u'_x v}{c^2}) + (u'_x - v)\frac{v}{c^2}\right]dv}{\left(1 - \frac{u'_x v}{c^2}\right)^2}$$

$$du_x = \frac{\frac{du'_x}{\gamma^2} - \left(1 - \frac{u'^2_x}{c^2}\right)dv}{\left(1 - \frac{u'_x v}{c^2}\right)^2}$$

$$cdt = \gamma(cdt' - \beta dx')$$

$$\rightarrow dt = \gamma dt' \left(1 - \beta \frac{u'_x}{c}\right)$$

$\therefore$

$$a_x = \frac{du_x}{dt} = \left[\frac{du'_x}{\gamma^2 \left(1 - \beta \frac{u'_x}{c}\right)^2} \right] \left[\frac{1}{\gamma dt' \left(1 - \beta \frac{u'_x}{c}\right)} \right] = \frac{a'_x}{\left[\gamma \left(1 - \beta \frac{u'_x}{c}\right) \right]^3}$$

$$\text{If } u'_x = v$$

$$\underline{a_x = \gamma^3 a'_x}$$

Note: Both S & S' frame are moving with same uniform acceleration.

Motion in Space-time diagram

$$\gamma^3 a'_x = \frac{\gamma a'_x}{1 - \beta^2} = \gamma a'_x (1 + \gamma^2 \beta^2)$$

γ is not a constant

$$\gamma = (1 - \beta^2)^{-1/2}$$

$$\frac{d\gamma}{dt'} = \frac{-1/2}{(1 - \beta^2)^{3/2}} \left(-2\beta \frac{dv}{dt'} \right) = \frac{1}{c^2 (1 - \frac{v^2}{c^2})^{3/2}} v a'_x = \frac{\gamma^3 v}{c^2} a'_x$$

$$v \frac{d\gamma}{dt'} = \gamma^3 \beta^2 a'_x$$

$$\therefore \gamma^3 a'_x = \gamma a'_x + v \frac{d\gamma}{dt'} = \gamma \frac{dv}{dt'} + v \frac{d\gamma}{dt'} = \frac{d(\gamma v)}{dt'}$$

$$\therefore a_x = \frac{d}{dt'} (\gamma v)$$

Since acc^{μ} is uniform.

$$\therefore \frac{v}{\sqrt{1 - v^2/c^2}} = a_x t'$$

$$\Rightarrow \boxed{v = \frac{a_x t'}{\sqrt{1 + \frac{a_x^2 t'^2}{c^2}}}}$$

Since $u'_x = v = \frac{dx'}{dt'}$

$$\Rightarrow \boxed{x'^2 - c^2 t'^2 = \frac{c^4}{a_x^2}} \rightarrow \text{Hyperbolic Trajectory.}$$

4 Acceleration

$$\frac{dU^\mu}{d\tau}$$

$$U^\mu U_\mu = c^2$$

$$\Rightarrow 2 \frac{dU^\mu}{d\tau} U_\mu = 0 \Rightarrow A^\mu U_\mu = 0 \Rightarrow A^\mu \perp U_\mu$$

$$A^\mu = \frac{dU^\mu}{d\tau} = \frac{d}{d\tau} [r(c, \vec{u})] = r [c \dot{\gamma}, \dot{\gamma} u + \gamma \dot{u}]$$

$$A^\mu A_\mu = c^2 \gamma^2 \dot{\gamma}^2 - \gamma^2 \dot{\gamma}^2 u^2 - 2 \gamma^3 \dot{\gamma} u \dot{u} - \gamma^4 \dot{u}^2$$

$$u = v$$

Show That:

- ① $\gamma v = c(r^2 - 1)^{1/2} \Rightarrow r = \frac{1}{\sqrt{1 - v^2/c^2}} \Rightarrow v = \frac{c}{\gamma} \sqrt{\gamma^2 - 1}$
- ② $c^2 dr = r^3 \vec{v} \cdot d\vec{v} \Rightarrow dr = \left(-\frac{1}{2} r\right) \left(-\frac{2 \vec{v} \cdot d\vec{v}}{c^2}\right) = \frac{r^3}{c^2} \vec{v} \cdot d\vec{v}$
- ③ $d(\gamma u) = \gamma^3 dv$

$$\therefore A^\mu A_\mu = -a^2 \gamma^4 - \frac{\gamma^6 (\vec{v} \cdot \vec{a})^2}{c^2}$$

$$= -\gamma^4 a^2 \left(1 + \frac{v^2}{c^2}\right)$$

$$\frac{v^2}{c^2} = \frac{v^2}{c^2}$$

$$\boxed{A^\mu A_\mu = -\gamma^4 a^2}$$

Resulto

$A^\mu A_\mu \Rightarrow$ Invariant Since γ changes $\Rightarrow$ 'a' must change

$$\gamma^3 a = a \Rightarrow \text{Invariant}$$

$$\Rightarrow \gamma^3 dv = a dt$$

Assume @ $t_0=0, u_0=0$

$$\Rightarrow v^2(t) = \left(1 - \frac{v^2}{c^2}\right) a^2 t^2$$

$$\boxed{\left(x + \frac{c^2}{a}\right)^2 - c^2 t^2 = \frac{c^4}{a^2}}$$

Hyperbola: We meet again ~~as~~
 This time I am more powerful,
 more omnipotent, more
 invariant to change. Ha Ha Ha
 Ha Ha.

Lecture 13 (5 Mar)

[Missed Lecture 12]
on 2 Mar

Note:

Laws of Physics are same in all freely falling locally inertial frames.

HW

Under coordinate transformation: $x^i \rightarrow \tilde{x}^k$

$$\tilde{\Gamma}_{kl}^i = \frac{\partial \tilde{x}^i}{\partial x^m} \frac{\partial x^n}{\partial \tilde{x}^k} \frac{\partial x^p}{\partial \tilde{x}^l} \Gamma_{np}^m + \frac{\partial^2 x^p}{\partial \tilde{x}^k \partial \tilde{x}^l} \frac{\partial \tilde{x}^i}{\partial x^p}$$

Locally Inertial Coordinates -

Coordinate x^μ - $\Gamma_s \neq 0$ @ arbitrary point P.

Construct

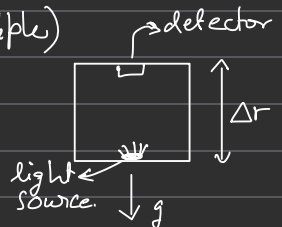
$$\tilde{x}^\nu = -\frac{1}{2} \Gamma_{\sigma\mu}^\nu (x^\mu - x_P^\mu) (x^\sigma - x_P^\sigma)$$

$$\tilde{x}_P^\nu = 0 \quad \left. \frac{\partial \tilde{x}^\nu}{\partial x^\mu} \right|_P = 0 \quad \left. \frac{\partial^2 \tilde{x}^\nu}{\partial x^\mu \partial x^\sigma} \right|_P = -\Gamma_{\mu\sigma}^\nu \longrightarrow \tilde{\Gamma} = 0$$

Consequences of SEP (Strong Equivalence Principle)

Freely falling bodies can be considered as Locally Inertial frames.

$$\therefore \Delta t = \frac{\Delta r}{c}$$



By the time the light reaches the detector, the frame has gained an additional velocity v due to gravity.

$$v = g \Delta t = g \frac{\Delta r}{c}$$

Initially, at source, light had a frequency ν . Due to gain in velocity, after reaching the detector, the frequency is changed to ν' . [Doppler Effect]

$$\frac{\Delta \nu}{\nu} \approx -\frac{g \Delta r}{c^2} : \text{Light gets redshifted} \quad (\nu' < \nu)$$

Assume a coordinate system where location of lift is given by r .

$t(r)$ = Time-Interval b/w ticks of a clock at rest at r .

$\nu(r)$ = frequency of clock at rest at r .

$\therefore$

$$t(r) = \frac{1}{\nu(r)}$$

$$t(r + \Delta r) = \frac{1}{\nu'} = \frac{1}{\nu - \Delta \nu} \approx \frac{1}{\nu} \left(1 + \frac{\Delta \nu}{\nu} \right) = t(r) \left(1 + \frac{GM \Delta r}{r^2 c^2} \right)$$

$$\Rightarrow t(r) + \frac{dt}{dr} \Delta r = t(r) \left(1 + \frac{GM \Delta r}{r^2 c^2} \right)$$

$$\Rightarrow \int_{t(r)}^{t(\infty)} \frac{dt}{t} = \int_r^{\infty} \frac{GM}{r^2 c^2} dr \Rightarrow \ln \left(\frac{t(\infty)}{t(r)} \right) = \frac{GM}{rc^2} \Rightarrow \frac{t(r)}{t(\infty)} = e^{-\frac{GM}{rc^2}} \approx \left(1 - \frac{GM}{rc^2} \right)$$

$$\Rightarrow t(r) = t(\infty) \left(1 - \frac{GM}{rc^2} \right)$$

$$\text{Let } \phi = -\frac{GM}{r} \text{ (Gravitational Potential)} \Rightarrow t(\phi) = t(0) \left(1 + \frac{\phi}{c^2} \right) \quad (\phi < 0)$$

Note: $\rightarrow$ Exact form.

$$\Rightarrow \boxed{t(0) > t(\phi)} \rightarrow \text{Proper Time.}$$

$\downarrow$
Coordinate at Infinity.

$$\boxed{d\tau^2 = dt^2 \left(1 - \frac{2GM}{rc^2} \right)} \text{ (will be proved later)}$$

Result: Light will be redshifted in presence of Gravity.

Experimentally shown by Pound-Rebka Experiment

$$\hookrightarrow \Delta r = 22.6 \text{ m}$$

$$\left. \frac{\Delta \nu}{\nu} \right|_{\text{Theoretical}} = 2.46 \times 10^{-5}$$

$$\left. \frac{\Delta \nu}{\nu} \right|_{\text{Expt}} = 2.57 \pm 0.2 \times 10^{-5} \quad (\text{very close})$$

Note:

We cannot just consider 2 frames - one when light was at the source & other when light was at the detector - and then apply special relativity to get related frequencies because the frames are accelerating & thus are only instantaneously inertial. This leads to discrepancy in our results.

Uniformly Accelerated Frame

$$u^\alpha u_\alpha = -c^2 \quad a^\mu u_\mu = 0 \quad a^\mu g_\mu = \text{constant}$$

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2$$

$$x = \frac{c^2}{g} \left[\cosh\left(\frac{g\tilde{t}}{c}\right) - 1 \right] + \tilde{x} \cosh\left(\frac{g\tilde{t}}{c}\right)$$

↗ Parameter

$$t = \frac{c}{g} \left[\sinh\left(\frac{g\tilde{t}}{c}\right) \right] + \frac{\tilde{x}}{c} \sinh\left(\frac{g\tilde{t}}{c}\right)$$

$$y = \tilde{y}, \quad z = \tilde{z}$$

∴ Line Element in new coordinates :

$$ds^2 = c^2 dt^2 \left(1 + \frac{\tilde{x}g}{c}\right)^2 - d\tilde{x}^2 - d\tilde{y}^2 - d\tilde{z}^2$$

Origin @ $(\tilde{x}, \tilde{y}, \tilde{z})$

$$x = \frac{c^2}{g} \left[\cosh\left(\frac{g\tilde{t}}{c}\right) - 1 \right] \quad y=0, \quad z=0$$

$$t = \frac{c}{g} \left[\sinh\left(\frac{g\tilde{t}}{c}\right) \right]$$

$$u^\mu = \left(\frac{cdt}{d\tilde{t}}, \frac{dx}{d\tilde{t}}, \frac{dy}{d\tilde{t}}, \frac{dz}{d\tilde{t}} \right) = \left(c \cosh\left(\frac{g\tilde{t}}{c}\right), c \sinh\left(\frac{g\tilde{t}}{c}\right), 0, 0 \right)$$

$$a^\mu = \frac{du^\mu}{d\tilde{t}} = \left(g \sinh\left(\frac{g\tilde{t}}{c}\right), g \cosh\left(\frac{g\tilde{t}}{c}\right), 0, 0 \right)$$

$$\therefore u^\mu u_\mu = -c^2; \quad a^\mu a_\mu = -g^2; \quad a^\mu u_\mu = 0$$

Lecture 14 [9 March]

Recap:

① 3+1 spacetime geometry

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

② SEP $\Rightarrow$ LIF = Freely falling frame in the local g .

$\hookrightarrow$ Laws of Physics hold true by putting $g_{\mu\nu} = \eta_{\mu\nu}$ (Minkowski metric)

$\hookrightarrow$ There exist a coordinate transformation s.t. local geometry in new coordinates is flat $\Rightarrow \Gamma'^s = 0$

In LIF, $\Gamma^s = 0$

$\therefore$

$$R^\alpha_{\beta\gamma\delta, \omega} = \Gamma^\alpha_{\beta\delta, \gamma\omega} - \Gamma^\alpha_{\beta\gamma, \delta\omega}$$

Bianchi Identity:

$$-R_{\alpha\beta\gamma\delta, \omega} + R_{\alpha\beta\omega\delta, \gamma} + R_{\alpha\beta\omega\gamma, \delta} = 0$$

Proof:

$$R_{\alpha\beta\gamma\delta} = g_{\alpha\lambda} R^\lambda_{\beta\gamma\delta}$$

$$\Rightarrow R_{\alpha\beta\gamma\delta, \omega} = g_{\alpha\lambda, \omega} R^\lambda_{\beta\gamma\delta} + g_{\alpha\lambda} R^\lambda_{\beta\gamma\delta, \omega}$$

$$\text{Note: } g_{\alpha\lambda, \omega} = g_{\alpha, \omega} g_\lambda + g_{\alpha, \omega} g_\lambda = \Gamma^\pi_{\alpha\omega} g_{\pi\lambda} + \Gamma^\pi_{\lambda\omega} g_{\pi\alpha} = 0 \quad (\because \Gamma = 0)$$

$$\Rightarrow R_{\alpha\beta\gamma\delta, \omega} = g_{\alpha\lambda} (\Gamma^\lambda_{\beta\delta, \gamma\omega} - \Gamma^\lambda_{\beta\gamma, \delta\omega})$$

$$\text{Ily } R_{\alpha\beta\omega\delta, \gamma} = g_{\alpha\lambda} (\Gamma^\lambda_{\beta\omega, \gamma\delta} - \Gamma^\lambda_{\beta\gamma, \omega\delta})$$

$$\text{Ily } R_{\alpha\beta\omega\gamma, \delta} = g_{\alpha\lambda} (\Gamma^\lambda_{\beta\omega, \delta\gamma} - \Gamma^\lambda_{\beta\delta, \omega\gamma})$$

$$\therefore -R_{\alpha\beta\gamma\delta, \omega} + R_{\alpha\beta\omega\delta, \gamma} + R_{\alpha\beta\omega\gamma, \delta} = 0 \quad \text{H.P.}$$

Using Bianchi Identity -

$$g^{\alpha\delta} (-R_{\alpha\beta\gamma\delta,\omega} + R_{\alpha\beta\gamma\omega,\delta} + R_{\alpha\beta\omega\delta,\gamma}) = 0$$

$$\Rightarrow R_{\beta\gamma\delta,\omega} + g^{\alpha\delta} R_{\alpha\beta\gamma\omega,\delta} - R_{\beta\gamma\omega,\delta} = 0$$

$$\Rightarrow R_{\beta\gamma,\omega} + g^{\alpha\delta} R_{\alpha\beta\gamma\omega,\delta} - R_{\beta\omega,\gamma} = 0$$

$$g^{\beta\omega} (R_{\beta\gamma,\omega} + g^{\alpha\delta} R_{\alpha\beta\gamma\omega,\delta} - R_{\beta\omega,\gamma}) = 0$$

$$g^{\beta\omega} R_{\alpha\beta\gamma\omega,\delta} = g^{\beta\omega} R_{\beta\alpha\omega\gamma,\delta} = R_{\alpha\omega\gamma,\delta} = R_{\alpha\gamma,\delta}$$

$$\Rightarrow R_{\gamma,\omega} + R_{\gamma,\delta} - R_{\omega,\gamma} = 0$$

$$\Rightarrow R_{\gamma,\omega} + R_{\gamma,\omega} - R_{\gamma,\gamma} = 0$$

$$\Rightarrow 2R_{\gamma,\omega} - \delta_{\gamma}^{\omega} R_{,\omega} = 0$$

$$\Rightarrow \boxed{(R_{\gamma}^{\omega} - \frac{1}{2}\delta_{\gamma}^{\omega} R)_{,\omega} = 0}$$

Put $\boxed{G_{\gamma}^{\omega} = R_{\gamma}^{\omega} - \frac{1}{2}\delta_{\gamma}^{\omega} R}$ Einstein Tensor

$$\therefore \underline{G_{\gamma,\omega} = 0}$$

Note: Einstein's Field Equation: $G^{\mu\nu} = kT^{\mu\nu}$

$T^{\mu\nu}$: Stress-Energy Tensor

Note: $G_{\mu\nu}$ is Symmetric.

$$G_{\mu\nu} = g_{\mu\omega} G_{\nu}^{\omega} = g_{\mu\omega} R_{\nu}^{\omega} - \frac{1}{2} g_{\mu\omega} \delta_{\nu}^{\omega} R = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R$$

Since $R_{\mu\nu}$ & $g_{\mu\nu}$ are both symmetric, $\therefore G_{\mu\nu}$ is symmetric.

Weak Field, Slow Motion Assumption

* Weak Field $\Rightarrow$ Spacetime is almost flat
gravitational field is very weak
"Linearised Gravity"

$$\therefore \boxed{g_{\mu\nu} \simeq \eta_{\mu\nu} + h_{\mu\nu}}$$

$\eta_{\mu\nu}$: Minkowski Metric

$h_{\mu\nu}$: Gravitational Field Metric
(Linear Perturbation).

* Slow Motion $\Rightarrow v \ll c$.

eg. of weak field, slow motion system -

Earth's Motion around the Sun. ($v = 30 \text{ km/s} \ll c$)

Newton's Law of Gravity

$$\nabla^2 \phi = 4\pi G \rho \quad (\phi = -\frac{GM}{r})$$

Laplacian ∇^2

To find K in $G^{\mu\nu} = K T^{\mu\nu}$, we need to match it with Newton's Gravity by incorporation weak field, slow motion assumption.

$$\begin{aligned} \Gamma_{\beta\gamma}^{\alpha} &= \frac{1}{2} g^{\alpha\delta} (g_{\alpha\beta,\gamma} + g_{\alpha\gamma,\beta} - g_{\beta\gamma,\alpha}) \\ &\simeq \frac{1}{2} (\eta^{\alpha\delta} + h^{\alpha\delta}) (h_{\alpha\beta,\gamma} + h_{\alpha\gamma,\beta} - h_{\beta\gamma,\alpha}) \end{aligned}$$

$$\Rightarrow \Gamma_{\beta\gamma}^{\alpha} = \frac{1}{2} \eta^{\alpha\delta} (h_{\alpha\beta,\gamma} + h_{\alpha\gamma,\beta} - h_{\beta\gamma,\alpha}) \quad (\text{Removing higher orders of } h)$$

$$\therefore \Gamma_{\beta\gamma}^{\alpha} = O(h)$$

$$\begin{aligned}
 R_{\mu\nu\sigma\omega} &= \frac{1}{2} (g_{\mu\omega, \nu\sigma} + g_{\nu\sigma, \mu\omega} - g_{\mu\sigma, \nu\omega} - g_{\nu\omega, \mu\sigma}) \\
 &\quad + g_{\alpha\beta} (\Gamma_{\nu\sigma}^{\alpha} \Gamma_{\mu\omega}^{\beta} - \Gamma_{\nu\omega}^{\alpha} \Gamma_{\mu\sigma}^{\beta}) \\
 &= \frac{1}{2} (h_{\mu\omega, \nu\sigma} + h_{\nu\sigma, \mu\omega} - h_{\mu\sigma, \nu\omega} - h_{\nu\omega, \mu\sigma}) + O(h^2)
 \end{aligned}$$

$\therefore$

$$R_{\nu\omega} = g^{\sigma\mu} R_{\mu\nu\sigma\omega} = \eta^{\sigma\mu} R_{\mu\nu\sigma\omega}$$

$$R_{00} = \eta^{\mu\sigma} R_{\mu 0 \sigma 0} = -\frac{1}{2} \eta^{\mu\sigma} h_{00, \mu\sigma} \simeq \frac{1}{2} \nabla^2 h_{00}$$

($\because$ It is a slow motion, $\therefore$ terms with time derivatives = 0)

From Einstein's Field Equation,

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = K T_{\mu\nu}$$

$$\Rightarrow g^{\mu\nu} R_{\mu\nu} - \frac{1}{2} g^{\mu\nu} g_{\mu\nu} R = K g^{\mu\nu} T_{\mu\nu}$$

$$\Rightarrow R = -KT \quad (T = g^{\mu\nu} T_{\mu\nu})$$

$\therefore$

$$R_{\mu\nu} = K (T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T)$$

$$\text{Thus, } R_{00} = K (T_{00} - \frac{1}{2} g_{00} T) = K (\rho c^2 - \frac{1}{2} \rho c^2) = \frac{1}{2} K \rho c^2$$

$$\Rightarrow \nabla^2 h_{00} = K \rho c^2$$

$$\text{Geodesic Eqn: } \frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\sigma}^{\mu} \frac{dx^\nu}{d\tau} \frac{dx^\sigma}{d\tau} = 0$$

Due to Slow Motion $\Rightarrow$ Non-Relativistic $\Rightarrow \tau = t$

$$\therefore \frac{d^2 x^\mu}{dt^2} + \Gamma_{\nu\sigma}^\mu \frac{dx^\nu}{dt} \frac{dx^\sigma}{dt} = 0$$

$$\frac{dx^i}{dt} = v_i, \quad \frac{dx^0}{dt} = c$$

Note: $(\nu, \sigma) = (0, 0)$ term will dominate here. ($\because v \ll c$)

$$\therefore \frac{d^2 x^\mu}{dt^2} + \Gamma_{00}^\mu c^2 = 0$$

$$\Gamma_{00}^\mu = \frac{1}{2} \eta^{\mu\nu} (2h_{\nu 0,0} - h_{00,\nu}) = -\frac{1}{2} \eta^{\mu\nu} h_{00,\nu}$$

$$h_{00} = -\frac{2GM}{rc^2} = \frac{2\phi}{c^2} \quad \left(\phi = -\frac{GM}{r} \right)$$

$$\therefore \Gamma_{00}^i = -\frac{1}{2} \eta^{ii} \left(\frac{2\phi}{c^2} \right)_{,i} = \frac{1}{c^2} \frac{\partial \phi}{\partial x^i}$$

$$\Rightarrow \frac{d^2 \vec{x}}{dt^2} = -\nabla \phi \Rightarrow \nabla^2 h_{00} = \frac{8\pi G \rho}{c^2} \quad \& \quad \nabla^2 h_{00} = K \rho c^2$$

$$\Rightarrow K = \frac{8\pi G}{c^4}$$

So.

$$\boxed{G^{\mu\nu} = \frac{8\pi G}{c^4} T^{\mu\nu}}$$

	Property.	Newtonian / Non-Relativistic	General Relativity
1.	Gravity Potential	ϕ	$g_{\mu\nu} = 1 + h_{\mu\nu}$
2.	Force	$-\nabla \phi$	$\Gamma_{\sigma}^{\mu}, g_{\mu\nu,\sigma}$
3.	E.O.M.	$\frac{d^2 \vec{x}}{dt^2} = -\nabla \phi$	$\frac{d^2 x^\mu}{dt^2} + \Gamma_{\nu\sigma}^\mu \frac{dx^\nu}{dt} \frac{dx^\sigma}{dt} = 0$
4.	Tidal Force	$\frac{\partial^2 \phi}{\partial x^i \partial x^j}$	Geodesic Deviation (Riemannian Tensor)
5.	Field Equation	$\nabla^2 \phi = 0; \nabla^2 \phi = 4\pi G \rho$	$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$

Schwarzschild Metric (1916)

* Exact solution of Einstein's field Equation.

* Gives metric around a spherically symmetric space and Vacuum.

$$T_{\mu\nu} = 0 \text{ except @ } r=0$$

$$ds^2 = \left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 - \frac{1}{\left(1 - \frac{2GM}{rc^2}\right)} dr^2 - r^2 d\theta^2 - r^2 \sin^2 \theta d\phi^2$$

M = Total Mass density

Note:

① Spherically symmetric, static & exact solution of Einstein's Equation.

② $r = r_0 = \frac{2GM}{c^2}$: Blows up the metric. But it can be resolved by change of coordinates.

③ If radial motion of photons $\Rightarrow$ No θ, ϕ dependence.

$$\Rightarrow ds^2 = \left(1 - \frac{r_0}{r}\right) c^2 dt^2 - \frac{1}{\left(1 - \frac{r_0}{r}\right)} dr^2 = 0 \quad (\text{Null like separation})$$

$$\Rightarrow \frac{cdt}{dr} = \pm \frac{1}{1 - \frac{r_0}{r}}$$

light cannot cross $r=r_0$, physical problem (Not Integrable)

This is not a singularity ($r=0$ is the physical singularity)

$\therefore r=r_0$ is a boundary that prevents information travel between inside & outside regions [Event Horizon]

Geodesic Equation

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\sigma}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\sigma}{d\tau} = 0$$

$$\Rightarrow \text{Let } p^\mu = m \frac{dx^\mu}{d\tau}$$

$$\Rightarrow m \frac{dp^\mu}{d\tau} + \Gamma_{\alpha\sigma}^\mu p^\alpha p^\sigma = 0$$

$$\begin{aligned} m \frac{dp_\beta}{d\tau} &= m \frac{d(g_{\beta\mu} p^\mu)}{d\tau} = g_{\beta\mu,\lambda} p^\mu m \frac{dx^\lambda}{d\tau} + m g_{\beta\mu} \frac{dp^\mu}{d\tau} \\ &= g_{\beta\mu,\lambda} p^\mu p^\lambda - g_{\beta\mu} \Gamma_{\alpha\sigma}^\mu p^\alpha p^\sigma \\ &= (\Gamma_{\beta\lambda}^\rho g_{\rho\mu} + \Gamma_{\mu\lambda}^\rho g_{\rho\beta}) p^\mu p^\lambda - g_{\beta\mu} \Gamma_{\alpha\sigma}^\mu p^\alpha p^\sigma \\ &= \Gamma_{\beta\alpha}^\gamma p^\alpha p_\gamma + \Gamma_{\alpha\sigma}^\mu g_{\mu\beta} p^\alpha p^\sigma - g_{\beta\mu} \Gamma_{\alpha\sigma}^\mu p^\alpha p^\sigma \end{aligned}$$

$$\therefore \boxed{m \frac{dp_\beta}{d\tau} - \Gamma_{\beta\alpha}^\gamma p^\alpha p_\gamma = 0}$$

Using this, note that -

$$\begin{aligned} \Gamma_{\beta\alpha}^\gamma p^\alpha p_\gamma &= \frac{1}{2} g^{\gamma\rho} (g_{\rho\beta,\alpha} + g_{\rho\alpha,\beta} - g_{\alpha\beta,\rho}) p^\alpha p_\gamma \\ &= \frac{1}{2} (g_{\rho\beta,\alpha} + g_{\rho\alpha,\beta} - g_{\alpha\beta,\rho}) p^\alpha p^\rho \\ &= \frac{1}{2} g_{\alpha\rho,\beta} p^\alpha p^\rho \end{aligned}$$

$$\therefore m \frac{dp_\beta}{d\tau} - \frac{1}{2} g_{\alpha\rho,\beta} p^\alpha p^\rho = 0$$

This imply that if $g_{\alpha\beta}$ is independent of x^β , then

$$\frac{dp_\beta}{d\tau} = 0 \Rightarrow p_\beta = \text{constant}.$$

Let $\beta = \phi \Rightarrow p_\phi = \text{constant}$ (Angular Momentum Conservation)

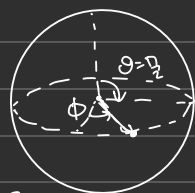
$$g_{\phi\phi} \dot{\phi} = r^2 \sin^2 \theta \dot{\phi} = \text{constant}.$$

(Independent of ϕ)

Consider $\theta = \frac{\pi}{2} \Rightarrow \underline{r^2 \dot{\phi} = \text{const.}}$ (Motion in Equatorial Plane)

Line element on Schwarzschild metric,

$$c^2 d\tau^2 = Z c^2 dt^2 - \frac{1}{Z} dr^2 - r^2 d\theta^2 - r^2 \sin^2 \theta d\phi^2 \quad \left(Z = 1 - \frac{2GM}{rc^2} \right)$$



$$\textcircled{a} \quad \theta = \frac{\pi}{2} \Rightarrow c^2 d\tau^2 = Z c^2 dt^2 - \frac{1}{Z} dr^2 - r^2 d\phi^2$$

$$\Rightarrow m^2 c^2 = m^2 c^2 Z \left(\frac{dt}{d\tau} \right)^2 - \frac{m^2}{Z} \left(\frac{dr}{d\tau} \right)^2 - m^2 r^2 \left(\frac{d\phi}{d\tau} \right)^2 \quad (\text{Constant})$$

Weak Field Limit $\Rightarrow Z \rightarrow 1$.

$$\therefore m^2 c^2 = m^2 c^2 \gamma^2 - (m^2 v_r^2 + m^2 r^2 v_\phi^2) \gamma^2$$

$$\Rightarrow m^2 c^2 = \frac{E^2}{c^2} - m^2 \gamma^2 (v_r^2 + r^2 v_\phi^2)$$

Variational principle -

$$\delta \int ds^2 = 0$$

$$\Rightarrow \delta \int (g_{\mu\nu} dx^\mu dx^\nu) = 0$$

$$\Rightarrow \oint \underbrace{\left(zc^2 \left(\frac{dt}{d\tau} \right)^2 - \frac{1}{z} \left(\frac{dr}{d\tau} \right)^2 - r^2 \left(\frac{d\phi}{d\tau} \right)^2 \right)}_L d\tau^2 = 0$$

$$\frac{\partial L}{\partial x^0} = \text{const} \quad , \quad \frac{\partial L}{\partial x^2} = \text{const} \quad , \quad \frac{\partial L}{\partial \left(\frac{d\phi}{d\tau} \right)} = \text{const} .$$

$$cP_0 = c g_{00} P^0 = m z c \frac{dx^0}{d\tau} = m z c^2 \frac{dt}{d\tau} = E .$$

$$P_\phi = g_{\phi\phi} P^\phi = \underbrace{r^2 \sin^2 \phi}_{\text{constant due to angular momentum conservation}} m \frac{d\phi}{d\tau} = mL \quad \left(\theta = \frac{\pi}{2} \right)$$

Lecture Notes From nptel

Special Relativity

Minkowski spacetime - $ds^2 = -c^2 dt^2 + d\vec{x} \cdot d\vec{x}$
 $= -(c^2 - v^2) dt^2$

Timelike separated events: $ds^2 < 0$

Spacelike separated events: $ds^2 > 0$

Lightlike separated events: $ds^2 = 0$

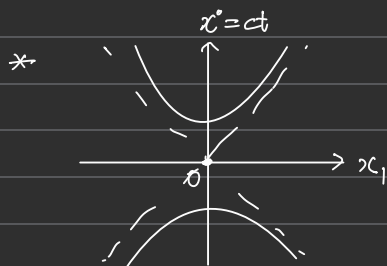
Timelike events: ($ds^2 < 0$)

An inertial ref. frame exists st. 2 events can happen at same location but different times.

S' : $d\vec{x}' = 0$, $dt' = d\tau \Rightarrow ds^2 = -c^2 d\tau^2 < 0$.

$-c^2 d\tau^2 = -c^2 dt^2 + d\vec{x} \cdot d\vec{x} \quad \therefore d\tau = \frac{dt}{\gamma} \quad \tau: \text{Proper time.}$

* For timelike separated events, time ordering b/w 2 events is Lorentz-invariant.



$$(x-0)^2 - c^2(t-0)^2 = -a^2$$

$$ct^2 - x^2 = a^2 : \text{Hyperbola.}$$

$$\frac{c^2 t^2}{x^2} = 1 + \frac{a^2}{x^2} = \text{Steepest.}$$

$$\lim_{x \rightarrow 0} \left(\frac{ct}{x} \right) = \pm 1 \Rightarrow \text{Asymptotes: } ct = \pm x.$$

Spacelike Events ($ds^2 > 0$)

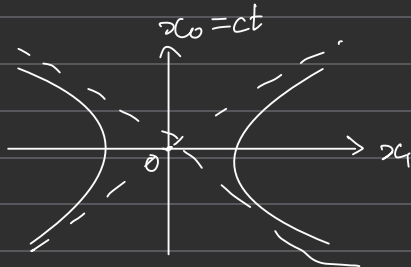
There exists an inertial ref. frame st. 2 events can happen at same time but at different locations.

S' : $dt' = 0$, $d\vec{x}' \neq 0 \Rightarrow ds^2 = (d\vec{x}')^2 > 0$

$dx' \rightarrow$ Proper distance.

$$ds^2 = dx'^2 = -cdt^2 + dx^2 \Rightarrow dx^2 = \underbrace{dx'^2}_{\text{Minimum separation possible}} + cdt^2$$

$\Rightarrow$ Length contraction.



$$-ct^2 + x^2 = x'^2$$

$\rightarrow$ The hyperbola represents a curve where the Lorentz invariant is same and constant w.r. any 2 inertial ref. frames
 $\hookrightarrow$ valid only for Flat spacetime curve (as in STR).

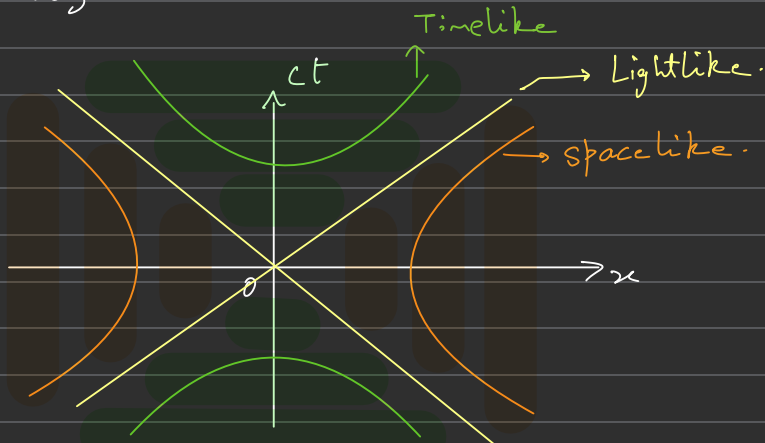
Null/ Lightlike events ($ds^2 = 0$)

2 Events that exist over a light's trajectory.

$$dx^2 = cdt^2 \Rightarrow ds^2 = 0.$$

$$\Rightarrow ct = \pm |x|$$

There is no rest frame where $dx' = 0$ or $dt' = 0$ b/w 2 events that are lightlike.



$$\text{Let } d\vec{x} = (dx^0, dx^1, dx^2, dx^3)^T \quad dx^0 = cdt$$

$$\therefore ds^2 = d\vec{x}^T \cdot \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot d\vec{x} \Rightarrow \text{Minkowski Metric.}$$

$$\eta = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \eta^T = \eta, \quad \eta^2 = I$$

Lorentz Transform: $x' = \Lambda x + a$

$$\Rightarrow dx' = \Lambda dx$$

$$\Rightarrow ds^2 = dx'^T \eta dx' = dx^T (\Lambda^T \eta \Lambda) dx = dx^T \eta dx$$

$$\therefore \boxed{\Lambda^T \eta \Lambda = \eta} \longrightarrow \text{Lorentz transforms preserves The Minkowski Metric.}$$

(Definition of Lorentz Transformation)

$\Lambda \rightarrow$ Pseudo-orthogonal matrix.

(If $\eta = I$, the Λ is orthogonal)

Lorentz Group:

Set of all Lorentz transforms Λ form a Lorentz Group.

① Closure Property:

$$\Lambda_1^T \eta \Lambda_1 = \eta, \quad \Lambda_2^T \eta \Lambda_2 = \eta$$

$$\Rightarrow (\Lambda_1 \Lambda_2)^T \eta (\Lambda_1 \Lambda_2) = \Lambda_2^T (\Lambda_1^T \eta \Lambda_1) \Lambda_2 = \Lambda_2^T \eta \Lambda_2 = \eta.$$

② Associative Property:

$$\Lambda_1 (\Lambda_2 \Lambda_3) = (\Lambda_1 \Lambda_2) \Lambda_3 \quad \because \text{Matrix multiplications are associative.}$$

③ Inverse:

$$\exists \Lambda^{-1} \text{ s.t. } \Lambda \Lambda^{-1} = \Lambda^{-1} \Lambda = I$$

$$\Lambda \eta \Lambda = \eta \Rightarrow \Lambda^T \Lambda^T \eta \Lambda \Lambda^{-1} = \Lambda^T \eta \Lambda^{-1} \Rightarrow \underline{\eta = \Lambda^T \eta \Lambda^{-1}}$$

④ Identity:

$$\exists I \text{ s.t. } \Lambda I = \Lambda; \quad I^T \eta I = \eta$$

We denote a Lorentz Group by $O(3,1)$

$\begin{matrix} \uparrow & \downarrow \\ 3 & 1 \\ +ve & -ve \\ \text{eigenvalues} & \text{eigenvalues} \end{matrix}$
of Minkowski spacetime.

$O(3,1)$: Group of all Pseudo-orthogonal Matrices.

Metric Preserving Groups

Given a metric g & a Group G , G preserves the metric g if —

$$\boxed{A^T g A = g}$$

$A \in G$ & A are linear, invertible transformations/matrix

If $g = \begin{bmatrix} -I_{q \times q} & 0 \\ 0 & I_{p \times p} \end{bmatrix} \Rightarrow G \leftarrow (p+q) \times (p+q)$ pseudo-orthogonal matrices.

$$O(p, q)$$

$\therefore$ Lorentz Groups are a special case of Metric preserving Groups with $p=3$, $q=1$.

Note: $g = \begin{pmatrix} 0 & I_{n \times n} \\ -I_{n \times n} & 0 \end{pmatrix}_{2n \times 2n} \rightarrow sp(n)$: Symplectic groups.

Going back to our Lorentz Groups —

$$\Lambda^T \eta \Lambda = \eta$$

$$\Rightarrow \det(\Lambda^T \eta \Lambda) = (\det \Lambda)^2 (\det \eta) = (\det \eta) \Rightarrow (\det \Lambda) = \pm 1.$$

If $\det \Lambda = +1 \Rightarrow$ Proper Lorentz Transformation.

If $\det \Lambda = -1 \Rightarrow$ Improper L.T.

$\det \Lambda = +1$ examples -

Note:

$$x' = \Lambda x$$

$$x^\mu = \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}$$

① Boost (B) $\Lambda(\beta_1, \beta_2, \beta_3)$ $\vec{\beta} = \frac{\vec{v}}{c}$

② Rotation (R) $\Lambda = \begin{pmatrix} 1 & \\ & O \end{pmatrix}$; $O^T O = I_{3 \times 3}$

Continuous
L.T.

$$\Lambda(\theta, \phi, \psi)$$

$\det \Lambda = -1$ examples -

① Parity (P) : Space Inversion (SI)

$$x'^0 = x^0, \quad \vec{x}' = -\vec{x}$$

$$\Lambda_P = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$$

② Time Reversal (T):

$$t' = -t$$

$$\vec{x}' = \vec{x}$$

$$\Lambda_T = \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$$

Discrete L.T.

Note: PT $\Rightarrow$ Proper L.T. (Strang Reflection)

Special Relativistic Mechanics

4-Vectors

3 Vector: $a = (a_x, a_y, a_z) = a^i e_i \quad i=1, 2, 3$ (Latin Indices)

4-Vector: $A = (a_t, a_x, a_y, a_z) = A^\mu e_\mu \quad \mu=0, 1, 2, 3$ (Greek Indices)

Length of a 4-vector = absolute value of spacetime distance between tail & tip of the vector.

Categorisation -

* Timelike 4-vector: Vector has timelike separation (Vector lies in timelike region).

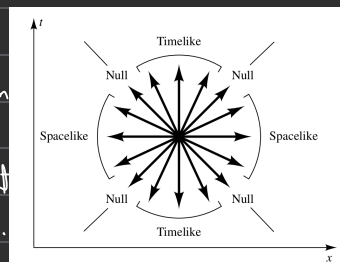
* Spacelike 4-vector: Vector has spacelike separation. (Vector lies in spacelike region)

* Nulllike 4-vector: Vector has nulllike separation. Such vectors have zero-length in spacetime.

All 4-Vectors are invariant quantities.

Their lengths & rules for addition/multiplication is same in all inertial frames.

But their components are different in different inertial frames as the basis vectors are different.



Note: Metric $g_{\alpha\beta} = e_\alpha \cdot e_\beta \Rightarrow a \cdot b = g_{\mu\nu} a^\mu b^\nu$

Let $u \Rightarrow$ Denote 3-vector.

$\underline{u} \Rightarrow$ Denote 4-vector.

4-Displacement

Let $A \equiv \underline{x}_A(ct_A, x_A, y_A, z_A)$ & $B \equiv \underline{x}_B(ct_B, x_B, y_B, z_B)$ be 2 events in spacetime.
 $\therefore$

Displacement 4-Vector: $\Delta \underline{x} = \underline{x}_B - \underline{x}_A$; $\Delta x^\alpha = x_B^\alpha - x_A^\alpha$

World Lines

- * Trajectory traced by a particle in spacetime is known as a worldline.
- * A particle follows a timelike worldline through spacetime.
- * A single parameter, say λ , can be used to parametrise the position of a particle over a world line.

$$x^\alpha = x^\alpha(\lambda) \quad \forall \alpha = 0, 1, 2, 3.$$

- * λ is typically chosen to be the proper time τ .

4-Velocity:

$$u^\alpha = \frac{dx^\alpha}{d\tau}$$

Let $\vec{v}$ be the 3-velocity.

$$\therefore v^i = \frac{dx^i}{dt} = \frac{dx^i/d\tau}{dt/d\tau} = \frac{1}{\gamma} \frac{dx^i}{d\tau} \Rightarrow \frac{dx^i}{d\tau} = \gamma v^i$$

$$\therefore \underline{u} = (\gamma c, \gamma \vec{v}) \quad : \text{4-Velocity in terms of 3-Velocity.}$$

4-Velocity represents the tangent along a worldline.

Note: $\underline{u} \cdot \underline{u} = -c^2$ in all reference frames.
(Invariant quantity)

4-Acceleration & 4-Force

$$\underline{a} = \frac{d\underline{u}}{d\tau} \quad \underline{f} = m \frac{d\underline{u}}{d\tau} = m \underline{a} \quad m: \text{rest mass.}$$

The equation $\underline{f} = m \underline{a}$ suggests 4 independent equations of motion. But there exists a constraint given by $\underline{u} \cdot \underline{u} = -c^2$ which reduces the number of independent equations to 3, same as in Newtonian Mechanics.

$$\frac{d(\underline{u} \cdot \underline{u})}{d\tau} = 2 \underline{u} \cdot \frac{d\underline{u}}{d\tau} = 0 \Rightarrow \boxed{\underline{f} \cdot \underline{u} = 0}$$

$$\frac{d\underline{u}}{d\tau} = \frac{\gamma^3 \vec{v}}{c^2} \frac{d\vec{v}}{d\tau} = \frac{\gamma^4}{c^2} \vec{v} \cdot \vec{a}$$

$$\therefore \underline{a} = \left(\frac{\gamma^4}{c} \vec{v} \cdot \vec{a}, \left(\frac{\gamma^4}{c^2} \vec{v} \cdot \vec{a} \right) \vec{v} + \gamma^2 \vec{a} \right)$$

Note: Proper acceleration = $\alpha = \sqrt{\underline{a} \cdot \underline{a}}$

for 1D Motion,

$$\alpha^2 = - \frac{\gamma^4 v^2 a^2}{c^2} + \frac{\gamma^4 v^2 a^2}{c^2} v^2 + \gamma^4 a^2 + \frac{2\gamma^6 v^2 a^2}{c^2} = \gamma^6 a^2$$

$$\therefore \boxed{\alpha = \gamma^3 |\vec{a}|}$$

Energy-Momentum 4-vector (4-Momentum)

$$\underline{p} = m \underline{u} = (\gamma m c, \gamma m \vec{v}) = \left(\frac{E}{c}, \vec{p} \right) \quad \begin{array}{l} E: \text{Energy.} \\ \vec{p}: \text{3-momentum.} \end{array}$$

$$\vec{p} = m \gamma \vec{v} = \frac{m \vec{v}}{\sqrt{1 - v^2/c^2}}$$

In the limit $\vec{v}^2 \ll c^2$

$$\Rightarrow \vec{p} \simeq m \vec{v} + O(v^3/c^2) \quad \left[\begin{array}{l} \text{Same as Newtonian} \\ \text{Momentum} \end{array} \right]$$

$$\frac{E}{c} = \frac{mc}{\sqrt{1-v^2/c^2}} \Rightarrow E = \frac{mc^2}{\sqrt{1-v^2/c^2}}$$

$$\text{In the limit } v^2 \ll c^2 \Rightarrow E \simeq c^2 \left(m + \frac{1}{2} m \frac{v^2}{c^2} \right) + \dots$$

$\downarrow$ Rest Mass $\rightarrow$ Kinetic Energy Term.

Since p^0 incorporates energy like terms, hence we named it E .

Note: $\boxed{p \cdot p = -m^2 c^2}$ (Invariant)

$$\therefore \boxed{E = c \sqrt{m^2 c^2 + p^2}}$$

$$\therefore \text{Rest Energy } (v=0) \Rightarrow \boxed{E = mc^2}$$

Since 3-Force $\vec{F} = \frac{d\vec{p}}{dt}$

$$\therefore \frac{d\vec{p}}{dt} = \gamma \vec{F}, \quad \underline{f} \cdot \underline{u} = 0 \Rightarrow -f^0 \gamma c + \gamma \vec{F} \cdot \gamma \vec{v} = 0 \Rightarrow f^0 = \gamma \vec{F} \cdot \frac{\vec{v}}{c}$$

Hence, $\boxed{\underline{f} = \left(\gamma \vec{F} \cdot \frac{\vec{v}}{c}, \gamma \vec{F} \right)}$

$$\text{Since } \frac{1}{c} \frac{dE}{dt} = f^0 = \gamma \vec{F} \cdot \frac{\vec{v}}{c} \Rightarrow \boxed{\frac{dE}{dt} = \vec{F} \cdot \vec{v}} \quad \left(\text{Power in Newtonian Mechanics} \right)$$

Rapidity

θ is known as rapidity & helps simplify calculations.

① $v = c \tanh \theta$

② $\gamma = \cosh \theta$

Variation principle for free particle motion

Just like in Newtonian mechanics, extremising action gives us the equilibrium path a system will take, here in Special Relativity, the worldline of a free particle is decided by extremising (maximising) proper time.

"The worldline of a free particle between 2 timelike separated points extremises the proper time between them."

$$\text{For a free particle} \Rightarrow \frac{d^2 x^\alpha}{d\tau^2} = 0 \quad [\text{No acceleration}]$$